

INTRAHOUSEHOLD INEQUALITY AND THE JOINT TAXATION OF HOUSEHOLD EARNINGS*

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Abstract

We derive the optimal joint-income tax schedule for couples, focusing on the distinction between interpersonal and inter-household inequality. Households are composed of two spouses with possibly unequal access to the family's economic resources. Individual-oriented utilitarianism typically leads to a misalignment between the households' and the government's objectives, a phenomenon termed *dissonance* by Apps and Rees (1988). The traditional ABC formula must be amended by including a Pigouvian term to correct for dissonance. Under general conditions, the effect of dissonance on marginal taxes is ambiguous; its sign depends on whether the less powerful spouse's marginal contribution to household earnings is less than, or greater than, her marginal entitlement to household consumption. Assuming identical iso-elastic preferences, the multidimensional heterogeneity collapses into a single-dimensional index, preserving the single-crossing property. This simplification enables us to solve Mirrlees (1971)'s multidimensional program and quantitatively assess the size and sign of the Pigouvian term, which is positive across all income levels, leading to higher marginal tax rates.

Keywords: *Intrahousehold Inequality; Joint Taxation; Collective Household JEL*

Classification: *H21, H31, D13.*

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1 Introduction

RESOURCES are not equally split across household members (Chavas et al. (2018); Chiappori and Meghir (2015); Haddad and Kanbur (1990)). This fact has significant and sometimes dramatic consequences for interpersonal and welfare inequality (Calvi (2020); Anderson and Ray (2010)). Despite these considerations, optimal tax theory often overlooks intrahousehold inequality concerns by adopting the unitary approach to household decisions. The richer alternative of collective models is rarely used because they inherently give rise to multidimensional screening problems, posing a significant challenge. In this paper, we place the intrahousehold distribution problem at center stage in a Mirrleesian context and explore its consequences for tax design.

Households in our model are composed of two spouses with different labor market productivities and utilities. Following the collective approach to household behavior (Chiappori (1988b, 1992); Apps and Rees (1988)), resources are allocated between spouses according to their relative power, captured by household-specific Pareto weights. If the primary goal of economic policies is to enhance individual well-being, tax policy evaluation should consider its impact on individuals rather than the household as a whole.¹

Our analysis focuses on joint income-splitting schedules. In this framework, the household problem is divided into two stages. In the first stage, for each level of household earnings and disposable income, spouses decide efficiently how to allocate effort and consumption. This solution is characterized by sharing rules, which map household earnings and disposable income to individual labor supply and consumption levels. These rules determine, based on preferences, power, and productivity, how much each spouse contributes to the household budget and how they share the total consumption.

Under the household's sharing rules, the second stage entails choosing household earnings in response to any given tax schedule. Earnings are chosen up to the point where further increases yield no additional benefit. These household decisions are shaped by power dynamics, captured by internal Pareto weights. The planner evaluates these decisions using a criterion that may differ from the household's, as relative power balances can diverge from the equal weighting a Utilitarian planner would assign. This misalignment between household and government objectives is termed *dissonance* by Apps and Rees (1988).²

Starting from the optimal Mirrleesian tax system, which disregards intrahousehold inequality, the existence of dissonance may motivate the planner to introduce additional

¹Neglecting the multi-person nature of households makes the typical welfare evaluation in optimal taxation literature incongruent with methodological individualism.

²This divergence between household objectives and planner objectives is similar to the 'decision utility' versus 'experience utility' tension in behavioral economics.

distortions in household earnings choices. Under dissonance, the planner is bound to be concerned with intrahousehold inequality and the internal dynamics of household decision-making due to a combination of two factors: the planner's welfare metric places more weight on the utility of the less-powered spouse, and the efficiency assumption, a cornerstone of collective models, implies that an increase in the utility of the least favored spouse results in a reduction in the utility of the other.

Although the planner only has joint tax instruments, by adjusting marginal tax rates, the planner can nudge household behavior towards more equitable choices, thereby reducing intrahousehold welfare inequality. Household inequality is specifically reduced whenever an increase (or decrease) in household earnings drives the spouse with lower bargaining power to extract, on the margin, more (or less) from the household budget than they contribute. This guarantees that the welfare gains from the tax system are distributed not only across but also within households, supporting social welfare goals that consider both inter and intra-household equity.

When acknowledging how dissonance alters tax policy design to address intrahousehold inequalities across the economy, we must reassess how the [Mirrlees \(1971\)](#) canonical program should be formulated. The challenge in employing the mechanism design approach to determine the optimal tax system in the [Mirrlees \(1971\)](#) problem is the inherent multidimensionality that arises from dealing with couples. We leverage the properties of iso-elastic preferences to reduce the dimensionality of family characteristics into a single-dimensional index, simplifying the mechanism design program into a single-dimensional screening problem where the single-crossing property still holds, as in [Heathcote and Tsujiyama \(2021\)](#). This approach enables us to maintain full generality with respect to the joint distribution of spouses' skills and Pareto weights, while still solving the mechanism design problem in terms of economic primitives. Furthermore, this functional form assumption offers a clear structure for the allocation of consumption and labor within the family: each spouse consumes a fixed share of the household consumption and commits to a proportional contribution to the household budget.

The novel element in our analysis, compared to [Heathcote and Tsujiyama \(2021\)](#), is the use of a Bergsonian utility function by households to assess their well-being. This family criterion assigns different Pareto weights to each spouse's utility, while the planner adopts an individual-oriented utilitarian criterion, aggregating spouses' well-being with equal weights to their utilities. The dissonance imposes, in the optimum, a Pigouvian adjustment to the traditional ABC formula for marginal tax rates, originally outlined by [Mirrlees \(1971\)](#) and further expanded by [Diamond \(1998\)](#); [Saez \(2001\)](#). The adjustment to incorporate the effects of dissonance results in an augmented formula, represented in

an ABC+D format. Our formula explicitly illustrates the importance of intrahousehold distribution, contrasting sharply with the prescriptions implied by any approach that treats households as singular units. Moreover, in our numeric exercise, the Pigouvian correction is always positive, leading to higher marginal tax rates than those obtained by ignoring dissonance.

Furthermore, we characterize the formulae for the optimal marginal tax rate using perturbation methods, without imposing any functional form restrictions on preferences. The multidimensionality of household types leads to heterogeneity in preferences at each earnings level. Implementing perturbations requires aggregating spouses' preferences and behavioral responses at each income level. The presence of dissonance introduces novel complexities in these aggregations, similar to those identified by [Gerritsen \(2016\)](#) in his work on the taxation of behavioral agents. In particular, we show that new and hard-to-estimate welfare-weighted versions of the usual behavioral elasticities are necessary for welfare evaluation (see equation (E.8) in the online Appendix E). In addition, we identify an additional sufficient statistic that must be considered: the wedge between the social value of income and its market price. Our findings, therefore, highlight that the assumption underlying perturbation methods, when applied to households, implicitly presumes a strong uniformity in elasticities across all families.

Finally, while our analysis centers on joint income-splitting tax schedules, we explore the welfare gains of gender-based subsidies in Section 6.³ We show that the utility of the less productive spouse may actually decrease after the introduction of a small subsidy on her earnings, resulting in a redistribution of utility towards the more productive spouse. When a spouse-specific income subsidy is introduced, it effectively taxes her leisure time, thereby increasing her labor supply. If the policy change does not affect power weights, the distribution of consumption remains unchanged; hence, the relative utility of the subsidized spouse may be reduced. When the least productive spouse is also the one with less power—as emphasized by [Immervoll et al. \(2011\)](#)—the optimal policy will typically introduce a small tax on the individual the policy aims to favor.⁴

³Despite a growing tendency towards individual taxes, joint tax schedules remain pervasive. The usual rationale for their use is based on principles of horizontal equity: couples with identical earnings should not be treated differently. Although we do not fully endorse their view, we acknowledge the compelling argument by [Gordon and Kopczuk \(2014\)](#), who find empirically that joint taxation performs better on equity grounds than separate taxation. In particular, individual earnings are a poor indicator of household earning ability. As a result, conditioning on other household-level information becomes useful when moving away from a separate tax schedule.

⁴This finding is also in line with [Cremer et al. \(2016\)](#). The result heavily depends on the assumption, commonly adopted in household taxation literature ([Kleven et al. \(2009\)](#); [Immervoll et al. \(2011\)](#); [Cremer et al. \(2016\)](#)), that household weights are invariant to perturbations in the policy. The presence of household production need not overturn this; in fact, it is more likely to reinforce the result.

The rest of the paper is organized as follows. After this introduction, we offer a brief literature review. Section 2 describes the economic environment. The policy objective is carefully discussed in Section 3. In Section 4, we solve for the optimal tax system using mechanism design under the assumption of identical iso-elastic preferences. Section 5 presents the quantitative exercises and explores the policy implications of our model. In Section 6, we partially generalize the environment to study the impacts of gender-dependent perturbation on the optimal joint tax schedule.

Related literature

Until recently, economists have approached household behavior based on a common preference framework, where all family members' resources were pooled to maximize a single objective function. Moreover, this 'household welfare function' held a normative status when applied in optimal tax theory (e.g., [Boskin and Sheshinski \(1983\)](#)). However, beginning with the pioneering contributions of [Manser and Brown \(1980\)](#) and [McElroy and Horney \(1981\)](#), and further advanced by the conceptual shift introduced by [Chiapori \(1988b, 1992\)](#); [Apps and Rees \(1988\)](#), Family Economics rapidly progressed toward the understanding that joint family decisions are driven by divergent interests within the household.⁵

[Boskin and Sheshinski \(1983\)](#) explore the sub-optimality of tax schedules like the one adopted in the US, where husbands and wives face equal marginal tax rates. They discuss the possibilities of taxing spouses equally or differently, or even subsidizing one of the spouses. Their analysis shows that taxing the earnings of husbands and wives at the same rate is inefficient because it disproportionately reduces female labor market participation. [Boskin and Sheshinski \(1983\)](#) adopt a unitary approach, using household (revealed) preferences as the normative criterion for the planner. Additionally, they limit their analysis to linear taxes. [Bastani \(2013\)](#) extends the analysis of linear taxes to a collective setting in which spouses make decisions through bargaining.

Also using a unitary approach, [Kleven et al. \(2009\)](#) analyze the general nonlinear optimal income tax for couples. In their model, the primary earner chooses labor supply along a continuum (intensive margin), while the secondary earner decides whether to participate in the labor market (extensive margin). If the secondary earner chooses to participate, their labor supply is fixed, whereas in our model, both spouses choose their labor supply along the intensive margin. [Kleven et al.](#) identify conditions under which the optimal tax scheme

⁵[Lundberg and Pollak \(1996\)](#), for example, adopted a bargaining protocol to model the independent agency of men and women within marriage. Early insights into the internal workings of families can also be found in many of Gary Becker's contributions

imposes a positive tax on secondary earnings and when those taxes on secondary earnings decrease as primary earnings increase. Their analysis begins with individual taxes and explores whether introducing jointness is optimal. By contrast, our analysis in Subsection 6.2 starts from a joint tax perspective and investigates whether differentiating marginal taxes can enhance welfare.

The multidimensional nature of household types poses well-known technical difficulties in a screening context (Rochet and Stole (2003)), which have slowed advances in household tax theory. As a consequence, most contributions to the literature limit their scope by pre-defining functional forms for taxes (Boskin and Sheshinski (1983)), precluding adjustments in all relevant decision margins (Kleven et al. (2009)), and/or restricting the degree of heterogeneity (Cremer et al. (2012)). Our assumption regarding household preferences and collective behavior introduces a form of dimensionality reduction, similar to the one exogenously imposed by Choné and Laroque (2010), while retaining the information necessary for assessing individual welfare. Under an income-splitting schedule, there is only one screening variable (household earnings) and three dimensions of characteristics (the two spouses' productivities and the bargaining weight parameter). Despite the bunching of different household types at each earnings level, these sets of agents are identified by a common exogenous parameter related to the disutility of generating earnings. Importantly, the exogeneity of this aggregated parameter to the policy allows for an exogenous ordering of households' willingness to exert effort, preserving single-crossing.⁶ A similar approach is used by Heathcote and Tsujiyama (2021) under the assumption of perfect insurance across large families, though their work does not consider the distribution of power and dissonance.

Studies by Immervoll et al. (2011), Cremer et al. (2012), and Cremer et al. (2016) are closely related to our work. Dissonance, as defined by Apps and Rees (1988), plays a central role in all of these studies. Immervoll et al. (2011) simplifies the analysis by focusing solely on extensive margin decisions and imposing strong restrictions on the choice sets. Cremer et al. (2012) handles multidimensionality by limiting couple types under the assumption of perfect assortative matching. Finally, Cremer et al. (2016) considers a finite number of types, allowing them to explore fully general taxes and demonstrate how tax formulae differ when dissonance is present. The limited number of types in their analysis enables a focus on a scenario where incentive constraints bind only in the conventional direction.

⁶For general preferences the ordering may be endogenous to the tax system, making the screening problem intractable without further restrictions. For the general case, we settle for a local characterization via perturbation methods, which we present as an extension in Subsection 6.1.

Guner et al. (2012) quantify the effects of tax reforms by carefully considering the labor supply of married women and the current demographic structure. They examine the situation where married women face heterogeneous costs of participation. Their analysis examines how the structure of taxation influences the participation decision, which is empirically sensitive to tax changes. In our setting, both men and women face heterogeneous participation costs, and couples do not have the option to file separately if both spouses are working. Our model is static, so we refrain from discussing important issues related to the interaction between risk sharing and spouses' labor supply analyzed in Blundell et al. (2016) and Wu and Krueger (2021). However, similar to these studies, joint labor supply decisions are central to our analysis.

Finally, it is important to mention Gerritsen's (2016) study on the taxation of behavioral agents. By distinguishing 'decision utilities' from 'experience utilities', his work identifies a misalignment between individuals' and planners' objectives, similar to dissonance. Similar concerns about aggregation also influence his optimal tax formulae. This form of 'smooth bunching' is observed in Saez (2001) where single-crossing is not imposed. The novelty introduced by dissonance is that there are two different relevant aggregated elasticities: the usual average elasticities and a welfare-weighted elasticity.⁷ Government revenues depend on these average elasticities, calculated at each income level using the empirical distribution. However, elasticities, as a measure of behavioral responses, also play a crucial role because dissonance makes them directly relevant to welfare evaluation.⁸ In this case, the average elasticity is obtained by aggregating across households at each income level using a welfare-adjusted distribution, similar to the risk adjustment commonly used in finance. This complexity in aggregation also arises in Gerritsen's work.

2 Environment

The economy is inhabited by a continuum of households (or families) with a measure normalized to one. Each household is composed of two spouses indexed by $i \in \{a, b\}$, where a identifies the female spouse and b , the male spouse in the household. Each spouse is characterized by a type $w_i \in [\underline{w}, \bar{w}] \subset \mathbb{R}_+$ denoting individual labor market productivity (or wage rate). Technology is linear as one efficient unit of labor produces one unit of consumption good. Agents are paid their marginal productivity, which implies that the

⁷In Saez's words: "It is not necessary to assume that people earning the same income have the same elasticity; the relevant parameters are simply the average elasticities at given income levels" (Saez (2001), p. 210).

⁸In particular, the envelope arguments used to limit the welfare consequences of tax perturbations to the mechanical effect cannot be applied when dissonance is present.

labor income generated by spouse $i \in \{a, b\}$ with productivity w_i supplying l_i hours of work is given by $z_i = w_i l_i$. Individuals derive utility from the consumption of a private good $x_i \in \mathbb{R}_+$ and disutility from marketable labor supply $l_i \in \mathbb{R}_+$ resulting in a total utility given by $U_i(x_i, l_i)$, common to all individuals of each gender. Utility $U_i : \mathbb{R}_+^2 \rightarrow \mathbb{R}$, $i = a, b$, is assumed to be strictly quasi-concave, strictly increasing in consumption, strictly decreasing in labor supply and of class $\mathcal{C}^2$.

Household decision making Households are collective in the sense that spouses' consumption and labor supply choices always result in efficient outcomes, regardless of the bargaining protocol they use. Therefore, we can assume that a household whose members have productivities (w_a, w_b) make consumption and labor supply decisions $(x_i, l_i)_{i \in \{a, b\}}$ to maximize a Bergsonian household utility function of the form

$$\alpha U_a(x_a, l_a) + (1 - \alpha) U_b(x_b, l_b), \quad (1)$$

for some Pareto weight $\alpha \in (0, 1)$, reflecting the relative contribution of the female spouse to household welfare. We impose no restrictions on the relationship between the Pareto weight, α , and spouses' productivities, (w_a, w_b) . Given the previous structure, each household can be parameterized by a triple $(w_a, w_b, \alpha) \in \Lambda \equiv [\underline{w}, \bar{w}] \times [\underline{w}, \bar{w}] \times (0, 1)$. This triple will henceforth be referred to as the family type, denoted by $\iota \equiv (w_a, w_b, \alpha)$. For a fixed value of α , household choices satisfy all the properties to be rationalized by complete and transitive preferences. With some abuse, we shall refer to them as the *household preferences*.

Informationally feasible allocations We follow the household economics literature by assuming that only total household consumption $x \equiv x_a + x_b$ is observable outside the household; that is, consumption cannot be assigned to a specific spouse. Because of this assumption, goods cannot be conditioned on each spouse's consumption. Given this informational structure, bundles that can be associated with a household in any mechanism are of the form $(x, z_a, z_b) \in \mathbb{R}_+^3$.⁹ Moreover, in most of this paper (except for Subsection 6.2), we restrict our analysis to policies that do not distinguish the income generated by each of the spouses. In this case, feasible bundles are of the form $(x, z) \in \mathbb{R}_+^2$, where $z \equiv z_a + z_b$ is the total income of the household. We call them *aggregate bundles*.

⁹Equivalently, (x, z_a, z_b) is such an allocation if and only if $\phi(x, z_a, z_b) \leq 0$, for some function $\phi : \mathbb{R}_+^3 \rightarrow \mathbb{R}$. These are the only budget sets that can be designed to induce the desired allocation by a planner.

Tax system The tax system that we study only uses information on total household income $z = z_a + z_b$, that is, $\mathbb{T}(z_a, z_b) = T(z_a + z_b)$. The government taxes households using a nonlinear joint income tax schedule, $T : \mathbb{R}_+ \rightarrow \mathbb{R}$, assigning a tax liability for each possible level of family income. Call $\mathcal{T}_0$ the set of all such tax systems. The after-tax income is used in the family consumption $x = z - T(z)$.

Beyond its empirical relevance, studying joint tax schedules is interesting because the impact of the intrahousehold distribution of power can be isolated by comparison with the well-established findings from optimal tax theory for individuals. Why so many governments rely on income-splitting schedules is, of course, an interesting question in itself.^{10,11} It is a well-acknowledged fact that a tax system cannot simultaneously achieve progressivity, ensure identical treatment for families with equal earnings, and maintain neutrality regarding marriage decisions. Consequently, prioritizing among these objectives becomes necessary, leading different societies to focus on the values they find most significant in shaping tax policies – see [Mirrlees et al. \(2011\)](#) p. 66.

From a more pragmatic perspective, this restriction on the space of feasible mechanisms will be convenient in reducing the number of margins that a government has to take into account to align incentives in this multidimensional environment.¹² Without such an assumption, aggregation would not be possible – e.g., [Heathcote and Tsujiyama \(2021\)](#) – and we would have to settle for stronger restrictions on the space of preferences or household choices.

Family problem As we have argued before, an important characteristic of the household consumption data is that most consumption is not assignable in the sense that only x , not x_a and x_b , is observed by outsiders. Moreover, under the restriction that policy instruments are based on total household earnings, the budget constraint is invariant to the source of labor income, and we can define for a type $\iota = (w_a, w_b, \alpha)$ family a utility function $V :$

¹⁰The same is true for the polar opposite, separable schedules, that are nowhere found in their idealized form in theoretical work.

¹¹The 2016 OECD Family Database lists several countries that rely on a joint income tax system. Estonia (for married couples), France (families), Germany (optional, married couples), Ireland (optional, married couples), Luxembourg (spouses and partners), Norway (optional), Poland (optional, married couples), Portugal (families), Spain (optional, families), Switzerland (married couples), and the United States (optional - married couples). Naturally, our model presents an idealized interpretation of income-splitting, based on total household income rather than the individual earnings of each spouse. [Cremer et al. \(2012\)](#) provide conditions under which the government should tax couples jointly.

¹²[Mirrlees \(1971\)](#) addresses the problem of choosing a nonlinear tax system as a screening problem. It has long been recognized that screening problems in multidimensional environments are hard to handle because the absence of an exogenous ordering of willingness to pay or work precludes a simple characterization of the binding constraints. It is also well known that the optimum of these problems usually implies too much *pooling*; see [Rochet and Choné \(1998\)](#) and [Rochet and Stole \(1987\)](#).

$\mathbb{R}_+^2 \times \Lambda \rightarrow \mathbb{R}$ as

$$V(x, z, \boldsymbol{\iota}) \equiv \max_{(x_i, z_i)_{i=a,b} \in \mathbb{R}_+^4} \left\{ \alpha U_a(x_a, l_a) + (1 - \alpha) U_b(x_b, l_b) \right. \\ \left. \text{s.t. } x_a + x_b = x \quad \text{and} \quad l_a w_a + l_b w_b = z \right\}. \quad (2)$$

The utility function V reflects the collective nature of family decisions and incorporates the intrahousehold allocation decisions of how much each spouse contributes to family income and how resources are distributed between them.¹³ It represents the ordering of (x, z) bundles by a household composed of spouses with productivities w_a and w_b and Pareto weight α . From this formulation, it is easy to see that different spouses from families choosing the same (x, z) can achieve very different utility levels depending on their relative productivities and Pareto weights.

Two important assumptions are embedded in this formulation: α is exogenous to any policy, and the decision process within the family will always lead to an efficient outcome. This latter assumption places our approach in the realm of collective models (e.g., [Chiappori \(1988a\)](#) and [Browning and Chiappori \(1998\)](#)). As for the former, we could hold Pareto weights fixed for two plausible reasons. First, Pareto weights could be determined at the marriage stage, as in [Gayle and Shephard \(2019\)](#).¹⁴ Second, as in [da Costa and de Lima \(2019\)](#), the planner may have additional instruments to handle Pareto weights. In both cases, an interesting question we leave for future work is how policies influence the distribution of weights in the long run through marriage markets. Since intrahousehold transfers are not observed, any policy cannot directly affect the allocation of consumption between spouses. Hence, for any feasible policy, household preferences ordering can be represented by the family utility function V .

¹³If we do not restrict the instruments beyond the informational constraint imposed by the non-assignability of consumption, the household utility for a type $\boldsymbol{\iota}$ family with $w_a, w_b > 0$ would be

$$V(x, z_a, z_b, \boldsymbol{\iota}) \equiv \max_{(x_a, x_b) \in \mathbb{R}_+^2} \left\{ \alpha U_a \left(x_a, \frac{z_a}{w_a} \right) + (1 - \alpha) U_b \left(x_b, \frac{z_b}{w_b} \right) \quad \text{s.t. } x = x_a + x_b \right\}.$$

¹⁴We do not take into account the policy impact on household formation. That is, we assume that couples do not divorce in response to changes in policies, nor do we allow spouses to anticipate those changes at the moment of marriage. [Alm and Whittington \(1999\)](#) find that in the US, the impacts of the income tax on marriage decisions, even when statistically significant, are small.

3 The Planner's Problem

Family characteristics $\boldsymbol{\iota}$ are heterogeneous in the population and assumed to be private information of the family. Let $F(\boldsymbol{\iota})$ denote their joint cumulative distribution over the support Λ . The planner can use the tax policy $T : \mathbb{R}_+ \rightarrow \mathbb{R}$ to influence couples' aggregate decisions and spouses' allocations indirectly. We assume the planner is restricted to using joint tax schedules $T \in \mathcal{T}_0$, featuring income splitting.

3.1 The planner's objective

We have associated with each household $\boldsymbol{\iota}$ a utility function $V(x, z, \boldsymbol{\iota})$ representing its preferences ordering over aggregate bundles. Whether any welfare meaning should be attached to it is a different matter. As [Chiappori and Meghir \(2015\)](#) have pointed out, a welfare metric that does not take into account intrahousehold inequality is inconsistent with any welfare criteria based on individuals, as is the case for utilitarianism. Intra-household inequality should, under this view, be central to the study of redistribution policies.

We therefore assume that, while taking “household preferences” as given, the planner's objective may depend on the utilities of all individuals in a different fashion. In particular, we study an anonymous individual-oriented utilitarian objective, where the planner maximizes the average of all individuals' utilities regardless of their bargaining weights.

Dissonance Under the individual-oriented utilitarian objective, the planner weights equally the utility of both spouses in a given couple. Spouses, in contrast, decide using a different objective (1) captured by their Pareto weights α and $1 - \alpha$. The solution to (2) defines consumption $x_a(x, z, \boldsymbol{\iota})$ and income $z_a(x, z, \boldsymbol{\iota})$ choices for the wife as a selection from the argmax correspondence. We can also directly define the analogous functions $x_b(x, z, \boldsymbol{\iota}) = x - x_a(x, z, \boldsymbol{\iota})$ and $z_b(x, z, \boldsymbol{\iota}) = z - z_a(x, z, \boldsymbol{\iota})$ for the husband. Determining the allocation of resources within the household. Using this notation, the household welfare *as perceived by the planner* is

$$W(x, z, \boldsymbol{\iota}) \equiv \frac{1}{2} \{V_a(x, z, \boldsymbol{\iota}) + V_b(x, z, \boldsymbol{\iota})\}, \quad (3)$$

where

$$V_i(x, z, \boldsymbol{\iota}) \equiv U_i \left(x_i(x, z, \boldsymbol{\iota}), \frac{z_i(x, z, \boldsymbol{\iota})}{w_i} \right), \quad i = a, b. \quad (4)$$

The function $W : \mathbb{R}_+^2 \times \Lambda \rightarrow \mathbb{R}$ represents the planner's assessment of individuals' welfare in a household $\boldsymbol{\iota}$ when consuming the aggregate bundle (x, z) .

We can combine (3) and (4) to write

$$W(x, z, \boldsymbol{\iota}) = V(x, z, \boldsymbol{\iota}) + \left(\frac{1}{2} - \alpha\right) \left[U_a \left(x_a(x, z, \boldsymbol{\iota}), \frac{z_a(x, z, \boldsymbol{\iota})}{w_a} \right) - U_b \left(x_b(x, z, \boldsymbol{\iota}), \frac{z_b(x, z, \boldsymbol{\iota})}{w_b} \right) \right], \quad (5)$$

where $V : \mathbb{R}_+^2 \times \Lambda \rightarrow \mathbb{R}$ defined in (2) represents the household preferences.

This expression has a nice behavioral analog. The first term on the right-hand side of (5) is the spouses' aggregate utility under the household welfare metric (2). It is analogous to the *decision utility* of some behavioral economics models – see [Farhi and Gabaix \(2020\)](#) – whereas the planner's ordering of bundles, W , is analogous to the *experience utility* of behavioral economics. It only coincides with the household's ordering when both spouses have equal power within the household, that is, when $\alpha = 1/2$. The difference between these two, represented by the second term on the right-hand side of Equation (5), underscores the misalignment between the household's welfare metrics and the planner's evaluation of the spouses' well-being. This concept, which [Apps and Rees \(1988\)](#) termed *dissonance*, captures the divergence in welfare evaluations.

Let z denote the household choice of family income under a smooth tax schedule, $T \in \mathcal{T}_0$. We can in this case define the **dissonance index**, ξ , through the total derivative of W with respect to z ,

$$\begin{aligned} \frac{dW}{dz} &= \frac{\partial W(x, z, \boldsymbol{\iota})}{\partial x} [1 - T'(z)] + \frac{\partial W(x, z, \boldsymbol{\iota})}{\partial z} \\ &= [1 - \xi(x, z, \boldsymbol{\iota})] \frac{\partial W(x, z, \boldsymbol{\iota})}{\partial x} [1 - T'(z)], \end{aligned} \quad (6)$$

where

$$\xi(x, z, \boldsymbol{\iota}) \equiv \frac{\partial W(x, z, \boldsymbol{\iota}) / \partial z}{\partial W(x, z, \boldsymbol{\iota}) / \partial x} \left[\frac{\partial V(x, z, \boldsymbol{\iota}) / \partial z}{\partial V(x, z, \boldsymbol{\iota}) / \partial x} \right]^{-1}. \quad (7)$$

The variable ξ is the ratio of the marginal rates of substitution according to the planner's and the household's metrics. It is equal to one whenever $\alpha = 1/2$. As we will explore throughout the text, ξ encapsulates all the necessary information about dissonance, playing a crucial role as a sufficient statistic in computing welfare within the context of the optimal taxation problem.

3.2 The Effect of Dissonance on Optimal Policy

In this subsection, we explore how variations that discourage (or stimulate) household earnings z impact the welfare of each spouse in a given couple. This analysis sheds light on how the government can foster redistribution within households, even when constrained to joint income taxation. Specifically, we establish a general principle for determining, starting from optimal tax formulas for unitary households, the direction of deviations induced by dissonance. This principle holds under the sole imposition of separability on preferences, i.e., $U_i(x_i, l_i) = u_i(x_i) - h_i(l_i)$. Here, for $i \in \{a, b\}$, both functions u_i and h_i are assumed to be strictly concave and convex, respectively, strictly increasing and of class $\mathcal{C}^2$.

In this section we focus on an arbitrary family, ι . For the sake of illustration we assume that the bargaining weight of spouse a is smaller than that of partner b , i.e., $\alpha < 1/2$, and omit ι from the notation for brevity. In the next section, we extend these considerations to the whole economy to calculate the optimal marginal tax rate.

With separable preferences, we can independently solve the two family sub-problems that follow. First, we determine how to allocate consumption between spouses,

$$\mathcal{U}(x) \equiv \max_{x_a \in [0, x]} \{ \alpha u_a(x_a) + (1 - \alpha) u_b(x - x_a) \}. \quad (8)$$

Second, we establish the most efficient way to share the responsibility of generating z between spouses,

$$\mathcal{H}(z) \equiv \min_{z_a \in [0, z]} \left\{ \alpha h_a \left(\frac{z_a}{w_a} \right) + (1 - \alpha) h_b \left(\frac{z - z_a}{w_b} \right) \right\}. \quad (9)$$

With a bit of notation abuse, let $x_a(x)$ and $z_a(z)$ represent their respective solutions. In possession of these policy functions, we can define, for this ι -family, the planner's assessment of (i) the value of consumption, x ,

$$\mathcal{V}(x) \equiv \frac{1}{2} \{ u_a(x_a(x)) + u_b(x - x_a(x)) \},$$

and; (ii) the cost of producing earnings, z ,

$$\mathcal{K}(z) \equiv \frac{1}{2} \left\{ h_a \left(\frac{z_a(z)}{w_a} \right) + h_b \left(\frac{z - z_a(z)}{w_b} \right) \right\}.$$

We aim to answer the question: ‘From the planner’s perspective, what is the welfare impact of a slight increase in household earnings, given the household’s existing choices?’

To begin, let us compare the impact of a slight increase in aggregate consumption from both the planner's and the household's perspectives, that is, $\frac{\partial \mathcal{V}(x)}{\partial x} - \frac{\partial \mathcal{U}(x)}{\partial x}$. Using the envelope condition for (8), we have

$$\frac{\partial \mathcal{U}(x)}{\partial x} = (1 - \alpha)u'_b(x_b). \quad (10)$$

The envelope theorem cannot be invoked to compute the impact from the planner's perspective, that is, $\partial \mathcal{V}(x)/\partial x$. In fact, this expression is given by

$$\frac{\partial \mathcal{V}(x)}{\partial x} = \frac{1}{2} [u'_a(x_a) x'_a(x) + u'_b(x_b) (1 - x'_a(x))] = \frac{(1 - 2\alpha)x'_a(x) + \alpha}{2\alpha} u'_b(x_b), \quad (11)$$

where we used the first-order condition $-\alpha u'_a(x_a) = (1 - \alpha)u'_b(x_b)$ – of the household consumption allocation problem (8) to derive the rightmost expression. Combining (10) and (11), it is immediate to see that

$$\frac{\partial \mathcal{V}(x)}{\partial x} \geq \frac{\partial \mathcal{U}(x)}{\partial x} \iff x'_a(x) \geq \alpha.$$

Similarly, using the envelope theorem to perform analogous calculations for the household costs of producing earnings, we have

$$\frac{\partial \mathcal{H}(z)}{\partial z} = (1 - \alpha) \frac{h'_b(l_b)}{w_b}. \quad (12)$$

The planner's assessment of these costs is

$$\frac{\partial \mathcal{K}(z)}{\partial z} = \frac{1}{2} \left[h'_a(l_a) \frac{z'_a(z)}{w_a} + h'_b(l_b) \frac{1 - z'_a(z)}{w_b} \right] = \frac{(1 - 2\alpha)z'_a(z) + \alpha h'_b(l_b)}{2\alpha w_b}, \quad (13)$$

where the first-order condition of the household cost problem (9) – namely $\alpha h'_a(l_a)/w_a = (1 - \alpha)h'_b(l_b)/w_b$ – was used for the last equality.

Combining (12) and (13) we get

$$\frac{\partial \mathcal{K}(z)}{\partial z} \leq \frac{\partial \mathcal{H}(z)}{\partial z} \iff z'_a(x) \leq \alpha.$$

For the planner to place a higher value on a marginal unit of consumption than the household does, the marginal share of consumption allocated to the spouse a , $x'_a(x)$, must be greater than the bargaining weight of that spouse, α .

Conversely, for the planner to view the marginal disutility of a marginal increase in earnings as less significant than the household does, the marginal contribution from the

spouse a , $z'_a(x)$, must be less than α .

If both criteria are met, the planner would advocate for an increase in household earnings. A simple yet powerful characterization can be drawn: the potential gains from a marginal increase in household earnings hinge on whether the spouse a gains more in extra consumption, on the margin, than her marginal contribution to the household budget.¹⁵

While meeting these conditions *at the same time* is sufficient for the planner to desire households to distort their earnings beyond the [Mirrlees](#)' optimal level, it is not necessary. This idea is conveyed in Proposition 1 below.

Proposition 1. *The necessary and sufficient condition for $dW/dz \geq 0$ at the household optimum is*

$$x'_a(x) \geq z'_a(z).$$

This condition can be equivalently expressed as

$$\xi(x, z) \leq 1,$$

where $\xi(x, z)$ is defined in (7).

Proof. See Appendix A. □

Under any tax schedule, at the optimal choice, the household balances the marginal cost of generating additional earnings with the marginal benefit of increased consumption. Therefore, any potential additional gain from the planner's perspective must result from redistribution between spouses. This can manifest in two ways: either by transferring consumption from the spouse with a lower marginal utility of consumption to the one with a higher marginal utility, or by reallocating the responsibility of earnings generation from the spouse with a higher disutility of effort to the one with a lower disutility of effort.

To help build an intuition for Proposition 1, it is helpful to consider the following special cases.¹⁶ Recall that we assume $\alpha < 1/2$, indicating, that the planner desires to redistribute from spouse b to spouse a . First, assume that the spouse a is entirely unproductive, such that $w_a = 0$. At the optimum, her labor supply is always zero, meaning she does not contribute to the household budget. However, since bargaining weights are positive – $\alpha \in (0, 1)$ –, she still receives a share of the family budget for consumption.

¹⁵This condition analogously implies that spouse b will be disadvantaged if he contributes more than he extracts from the household budget. Therefore, if an increase in household earnings benefits the spouse a , it will necessarily disadvantage the spouse b . This result is a consequence of the efficiency assumption in collective models.

¹⁶We thank an anonymous referee for suggesting the use of these special cases to illustrate the content of our main proposition.

This implies that while she does not bear any burden of generating household earnings, z , she benefits from increased consumption. Therefore, spouse a gains from an increase in household earnings. In this scenario, the government would aim to stimulate household earnings, resulting in lower marginal tax rates.

Second, consider the other extreme case. Assume now that spouse b , the one with higher bargaining weights, is entirely unproductive, such that $w_b = 0$. In this case, the labor supply of spouse b is always zero, making spouse a the sole contributor to the household budget. Both spouses benefit from an increase in earnings. Therefore, at the household optimum, an increase in household earnings only hurts the spouse a . Consequently, the planner would want to discourage household earnings, leading to higher optimal marginal tax rates.

Finally, for illustration, consider the specific case where the spouses' utilities are identical and ln-isoelastic (i.e., $U_i(x_i, l_i) = \ln x_i - l_i^{1+\gamma}/(1+\gamma)$), and both spouses are equally productive so that $w_a = w_b$. Even though both spouses are equally productive and have the same Frisch elasticity, spouse a contributes more to household earnings due to her lower bargaining weight α . Concurrently, spouse a is entitled to a smaller share of household consumption than spouse b , again because of her lower bargaining weight ($\alpha < 1/2$).¹⁷ Thus, spouse a is disadvantaged by an increase in household earnings. As a result, the planner would aim to discourage household earnings, leading to higher optimal marginal tax rates.

Proposition 1 establishes the conditions on behavioral responses within the intrahousehold allocation problem that justify the planner to deviate from any household-based allocation, in particular, the allocation emerging from a Mirrleesian solution. These conditions are expressed in terms of the derivative of the household choices, $x_a(x)$ and $z_a(z)$, which are endogenous objects. Proposition 2 below translates the behavioral conditions from Proposition 1 into conditions *on primitives* under which such deviations are optimal. For this, let us introduce $\sigma_i(x_i)$ the coefficient of relative risk aversion for utility $u_i(x_i)$, and $\gamma_i(l_i)$ the inverse of the Frisch elasticity of the labor supply:

$$\sigma_i(x_i) \equiv \frac{-u_i''(x_i)x_i}{u_i'(x_i)} \quad \text{and} \quad \gamma_i(l_i) \equiv \frac{h_i''(l_i)l_i}{h_i'(l_i)}, \quad i \in \{a, b\}.$$

Proposition 2. *At the household optimum, an increase in household earnings (weakly)*

¹⁷In Section 4, we carefully solve the family problem in the ln-isoelastic case and demonstrate these features of the solution.

increases $W(x, z, \iota)$ if and only if

$$\frac{x_a/\sigma_a(x_a)}{z_a/\gamma_a(l_a)} \geq \frac{x_b/\sigma_b(x_b)}{z_b/\gamma_b(l_b)}.$$

Proof. See Appendix A. □

To prove Proposition 2, we apply optimal rules for consumption and earnings-sharing formalized in Lemma 2 in Appendix A.

Lemma 2 impliest that the efficient sharing of household consumption corresponds to Borch's (1962) rule that relates the marginal participation in total consumption to Pareto weights and the local curvature of the utility for consumption function. The efficient assignment of earnings requirements is fully determined by the Frisch elasticity of labor supply and the Pareto weights. In the special case of identical iso-elastic preferences, we obtain Corollary 1.

Corollary 1. *At the household optimum, an increase in household earnings (weakly) increases $W(x, z, \iota)$ if and only if*

$$\frac{w_a}{w_b} \leq \left(\frac{\alpha}{1 - \alpha} \right)^{\frac{\gamma + \sigma}{\sigma(\gamma + 1)}},$$

where $u_i(c) = (c^{1-\sigma} - 1)/(1 - \sigma)$ for $\sigma \neq 1$ and $u_i(c) = \ln c$ for $\sigma = 1$, for $i \in \{a, b\}$. Additionally $h_i(l) = l^{1+\gamma}/(1 + \gamma)$.

Corollary 1 reveals at its core that the planner aims to encourage an increase in household earnings if spouse b 's productivity is significantly higher than spouse a 's. Under ln-isoelastic preferences (i.e., $\sigma = 1$), the condition in Corollary 1 is satisfied when the ratio of female to male productivity falls below the ratio of their Pareto weights, $w_a/w_b < \alpha/(1 - \alpha)$. If $w_a = w_b$, we find ourselves in a previously discussed case. It is crucial to keep this result in mind when interpreting the quantitative findings of Section 5. The empirical relationship between these ratios is pivotal in determining the direction of the distortions the planner aims to introduce in an optimal tax system.

4 Optimal Taxation

In section 3.2, we discussed how distortions can induce welfare redistribution within a specific family. Building upon this, the current section expands the analysis to encompass the entire economy. We now explore how a government can strategically design a tax system that not only facilitates redistribution between households but also effectively achieves

redistribution within them. Under common isoelastic preferences, household preferences can be aggregated into similar isoelastic preferences with a one-dimensional index of heterogeneity, referred to as type. Preferences over consumption and earnings exhibit single-crossing regarding this type. This aggregation allows us to derive the optimal tax formula using [Mirrlees's](#) mechanism design approach. In the case of $\ln$ utility of consumption, this type corresponds to the relevant household productivity, denoted by $\omega(\boldsymbol{\iota})$, as we will explore further.

By the revelation principle, the planner's problem can be restated in the space of direct mechanisms as follows:

$$\max_{(x,z):\Lambda \rightarrow \mathbb{R}_+^2} \int_{\Lambda} W(x(\boldsymbol{\iota}), z(\boldsymbol{\iota}), \boldsymbol{\iota}) dF(\boldsymbol{\iota}),$$

subject to incentive compatibility constraints: for every $\boldsymbol{\iota} \in \Lambda$

$$\boldsymbol{\iota} \in \arg \max_{\tilde{\boldsymbol{\iota}} \in \Lambda} V(x(\tilde{\boldsymbol{\iota}}), z(\tilde{\boldsymbol{\iota}}), \boldsymbol{\iota}).$$

We slightly abused notation in using $x(\cdot)$ and $z(\cdot)$ to denote the outcome function associated with the direct mechanism and the resource constraint:

$$\int_{\Lambda} [z(\boldsymbol{\iota}) - x(\boldsymbol{\iota})] dF(\boldsymbol{\iota}) \geq 0.$$

In this section, we delve into the particular scenario outlined by Corollary 1 where $\sigma = 1$. Implying that both spouses have logarithmic utility over consumption and isoelastic labor disutility functions. Concretely, assume for each $i \in \{a, b\}$ that preferences are identical and of the form

$$U_i(x_i, l_i) = \ln x_i - \frac{l_i^{1+\gamma}}{1+\gamma}. \quad (14)$$

Under an income-splitting schedule, a highly convenient aggregation result emerges.¹⁸ In fact, the choices of a type $\boldsymbol{\iota} = (w_a, w_b, \alpha)$ household can be represented as if they were made by an individual with log-isoelastic preferences of the form

$$\ln x - \frac{1}{1+\gamma} \left(\frac{z}{\omega(\boldsymbol{\iota})} \right)^{1+\gamma}, \quad (15)$$

¹⁸Isoelastic preferences are in line with the empirical evidence reviewed in [Chetty \(2006\)](#) and it is one of the most often parametrizations for preferences used in the taxation literature (see [Tuomala \(2016\)](#)). In the Appendix B we consider the case $u(x) = (x^{1-\sigma} - 1)/(1 - \sigma)$ for $\sigma \neq 1$.

where

$$\omega(\boldsymbol{\iota}) \equiv \left[\alpha^{\frac{-1}{\gamma}} w_a^{\frac{1+\gamma}{\gamma}} + (1-\alpha)^{\frac{-1}{\gamma}} w_b^{\frac{1+\gamma}{\gamma}} \right]^{\frac{\gamma}{1+\gamma}}$$

is defined as the **household productivity**.

To see this, consider a household's least costly way of producing z , which involves optimizing the allocation and efforts among its members - see (9)

$$\mathcal{H}(z, \boldsymbol{\iota}) = \frac{1}{1+\gamma} \min_{z_a \in [0, z]} \left\{ \alpha \left(\frac{z_a}{w_a} \right)^{1+\gamma} + (1-\alpha) \left(\frac{z - z_a}{w_b} \right)^{1+\gamma} \right\}.$$

Each spouse's contribution to total household earnings, $k_i = z_i/z$, can be expressed in terms of their individual productivities, (w_a, w_b) , and the Pareto weight α as

$$k_a \equiv \frac{z_a}{z} = \alpha^{\frac{-1}{\gamma}} \frac{w_a^{\frac{1+\gamma}{\gamma}}}{\omega(\boldsymbol{\iota})^{\frac{1+\gamma}{\gamma}}} \quad \text{and} \quad k_b \equiv \frac{z_b}{z} = (1-\alpha)^{\frac{-1}{\gamma}} \frac{w_b^{\frac{1+\gamma}{\gamma}}}{\omega(\boldsymbol{\iota})^{\frac{1+\gamma}{\gamma}}}. \quad (16)$$

Substituting z_a and z_b into (4) gives us

$$\mathcal{H}(z, \boldsymbol{\iota}) = \frac{1}{1+\gamma} \left(\frac{z}{\omega(\boldsymbol{\iota})} \right)^{1+\gamma}. \quad (17)$$

In Subsection 3.2, we examined the marginal contribution of each spouse to household income. With the specification for preferences in this section, we can derive the total contribution in closed form. It inversely varies with the spouse's individual Pareto weight; both consumption and leisure increase with one's Pareto weight. Additionally, individual earnings positively vary with spousal relative productivity, indicating that more productive spouses work relatively more, holding all other primitives fixed.

The allocation of consumption between spouses, as we have seen, follows from a straightforward application of Borch's rule. In the case of ln preferences, this simplifies to $x'_a(x) = \alpha$. Thus, we have the household utility

$$V(x, z, \boldsymbol{\iota}) = \mathcal{U}(x, \boldsymbol{\iota}) - \mathcal{H}(z, \boldsymbol{\iota}) = \kappa(\alpha) + \ln x - \frac{1}{1+\gamma} \left(\frac{z}{\omega(\boldsymbol{\iota})} \right)^{1+\gamma}, \quad (18)$$

where $\kappa(\alpha) \equiv \alpha \ln \alpha + (1-\alpha) \ln(1-\alpha)$.

Two noteworthy features in (18). First, family characteristics influence choices solely through $\omega(\boldsymbol{\iota})$. Given $\omega(\boldsymbol{\iota})$, $\kappa(\alpha)$ has no bearing on household preferences over (x, z) bundles. Families with the same $\omega(\boldsymbol{\iota})$ share the same objective, making $\omega(\boldsymbol{\iota})$ a sufficient statistic for family characteristics when computing the value assigned to a given bundle

$(x, z) \in \mathbb{R}_{++} \times \mathbb{R}_+$. Second, marginal rates of substitution $h'(z/\omega(\iota))/u'(c)$ are exogenously ordered which enable us to follow [Mirrlees](#)' approach: screening households based on their unitary type $\omega(\iota)$.

The combination of isoelastic preferences for spouses and joint income-splitting taxes allows us to reduce the household's multi-dimensional type ι into a uni-dimensional aggregator, preserving single-crossing. Assuming the existence of such an aggregator allows [Choné and Laroque \(2010\)](#) to study taxation in a multi-dimensional setting. Here, we show that, as in [Heathcote et al. \(2017\)](#), household cooperative behavior implies the existence of such an aggregator. Note, however, that the value that the planner assigns to resources of a family ι is not summarized by ω , as is the case in [Choné and Laroque \(2010\)](#).

The chosen parametrization of preferences ensures that both spouses always supply a positive quantity of labor. Consequently, decisions are made exclusively along the intensive margin. In a prior version of the paper – [Alves et al. \(2021\)](#) – we explore the introduction of participation decisions in an extended model, incorporating an additional discrete participation cost. This extension provides a means to better approximate the empirical differences in labor supply elasticities between married men and women

4.1 Optimal tax formulae

We consider a government that taxes households using a nonlinear joint income tax schedule $T : \mathbb{R}_+ \rightarrow \mathbb{R}$ to maximize a utilitarian social welfare function. For every ω , let $\mathcal{I}(\omega) \equiv \{\iota \in \Lambda \mid \omega(\iota) = \omega\}$ be the set of all family types ι whose behavior is summarized by the same ω . As seen before, for any ι -household with $\iota \in \mathcal{I}(\omega)$, the optimization problem it faces under a tax schedule $T : \mathbb{R}_+ \rightarrow \mathbb{R}$ is

$$\max_{(x,z) \in \mathbb{R}_{++} \times \mathbb{R}_+} \left\{ \ln x - \frac{1}{1+\gamma} \left(\frac{z}{\omega} \right)^{1+\gamma}, \quad \text{s.t.} \quad x = z - T(z) \right\}. \quad (19)$$

Also, let us define the cumulative distribution of ω as

$$\Psi(\omega) = \int_{[\omega(\iota) \leq \omega]} dF(\iota) \quad (20)$$

and assume that its associated density $\psi(\omega)$ is well-defined and strictly positive everywhere.

This approach allows us to consider the planner's program as if it involved a representative individual representing the family. This individual derives utility from the aggregate bundle (x, z) , and its type is the one-dimensional parameter ω . Aggregate 'household preferences' are well-behaved and exhibit exogenously ordered behavior with respect to

the parameter ω (i.e., the *Spence-Mirrlees condition* is satisfied). The first-order necessary condition for the ω -family's problem (19) is

$$1 - T'(z) = xz^\gamma / \omega^{1+\gamma}.$$

To simplify the planner's program we start by defining, for all ω ,

$$\bar{W}(x, z, \omega) \equiv \mathbb{E} [W(x, z, \iota) | \iota \in \mathcal{I}(\omega)], \quad \text{and} \quad \bar{V}(x, z, \omega) \equiv \mathbb{E} [V(x, z, \iota) | \iota \in \mathcal{I}(\omega)],$$

where the expectations operator $\mathbb{E}$ takes cross-sectional averages. These functions represent ω -household average versions of the planner's and households' welfare assessments, respectively. If we average $\xi(x, z, \iota)$ across all $\iota \in \mathcal{I}(\omega)$ we define a ' ω -household average dissonance index',

$$\bar{\xi}(\omega) \equiv \mathbb{E} [\xi(x, z, \iota) | \iota \in \mathcal{I}(\omega)], \quad (21)$$

which we will refer to simply as the ω -household wedge, and note that

Lemma 1. *If preferences are represented by (14), then, for all ω ,*

$$\bar{W}(x, z, \omega) = \bar{V}(x, z, \omega) + [1 - \bar{\xi}(\omega)] \frac{1}{1 + \gamma} \left(\frac{z}{\omega} \right)^{1+\gamma}.$$

Proof. See Appendix A. □

Lemma 1 allows us to re-write the planner's objective as

$$\max_{(x(\omega), z(\omega))_\omega} \int_{\underline{\omega}}^{\bar{\omega}} \left\{ \bar{V}(x(\omega), z(\omega), \omega) + [1 - \bar{\xi}(\omega)] \frac{1}{1 + \gamma} \left(\frac{z(\omega)}{\omega} \right)^{1+\gamma} \right\} \psi(\omega) d\omega, \quad (22)$$

while keeping the incentive compatibility constraint¹⁹

$$\omega \in \operatorname{argmax}_{\hat{\omega}} V(x(\hat{\omega}), z(\hat{\omega}), \omega),$$

and the resource constraint

$$\int_{\underline{\omega}}^{\bar{\omega}} [z(\omega) - x(\omega)] \psi(\omega) d\omega \geq G.$$

These conditions are identical to those in the single-person program by [Mirrlees \(1971\)](#). Solving the planner's program leads us to the main proposition of this section.

¹⁹Note that all households ι with $\iota \in \mathcal{I}(\omega)$ rank (x, z) bundles the same way. From an incentive perspective the distinction between V and $\bar{V}$ is, therefore, inconsequential.

Proposition 3. *In all intervals where $z(\omega)$ is strictly increasing, the marginal tax rate for the optimal (nonlinear) joint income tax schedule is given by*

$$\frac{T'(z(\omega))}{1 - T'(z(\omega))} = \frac{1 + \gamma}{\psi(\omega)x(\omega)\omega} \int_{\omega}^{\bar{\omega}} [x(s) - \mathbb{E}[x]] \psi(s) ds - \frac{\mathbb{E}[x]}{x(\omega)} [1 - \bar{\xi}(\omega)]. \quad (23)$$

Proof. See Appendix A. □

Since the planner has access only to the joint tax system, he must use this instrument to achieve a triple task. By distorting allocations through the increase in marginal tax rates, the planner affects inefficiencies and implements redistribution both within and across households. As we have seen in Section 3.2, the planner can redistribute utility between spouses by distorting the family's choice of total earnings by using taxes to influence the family's decision at the margin. The second term on the right-hand side of equation (23) captures this motive. The sign of this Pigouvian correction term depends on whether $\bar{\xi}(\omega)$, for the reasons outlined in Proposition 1. The ratio between the marginal value of resources on the hands of ω households and the marginal cost of funds weighs on the importance of this distortion.

As for the first term on the right-hand side of equation (23), which we refer to as the Mirrleesian term, it captures the inter-families efficiency-equity trade-off and is the ln-isoelastic version of [Mirrlees' \(1971\)](#) formula.

5 Quantitative Assessment

We now compute the marginal tax rate derived in Proposition 3 using real data from the March 2016 extract of the CPS. Our goal with this exercise is to evaluate whether dissonance is potentially relevant from a quantitative perspective. Toward this end, we will assess how family decision-making affects the size and shape of optimal joint income taxes and the role of dissonance. In Appendix C, we have a detailed exposition of the numerical implementation.

5.1 Numerical Exercise

We restrict our sample to married spouses whose incomes are strictly positive. As in [Mankiw et al. \(2009\)](#), we use hourly wages as a proxy for productivity. We also use the hours worked per week as our labor supply measure. The parameter γ in the disutility of labor is set to be equal to 2, implying a Frisch elasticity of labor supply of 0.5, as in [Chetty \(2009\)](#). Given our assumptions, for a level of household resources z , the optimal allocation

(z_a, z_b) connects spouses' labor supplies — $l_i = z_i/w_i$, $i \in \{a, b\}$ — wages, and relative power as follows,

$$\frac{l_a}{l_b} = \left[\left(\frac{1-\alpha}{\alpha} \right) \left(\frac{w_a}{w_b} \right) \right]^{1/\gamma}. \quad (24)$$

Using actual data on wages, labor supply, and equation (24), we can uniquely pin down a value of α for each family, fully identifying the household type, $\boldsymbol{\iota} = (w_a, w_b, \alpha)$.²⁰ The implied unconditional empirical distribution of α is depicted in Figure 3 in the Appendix C. Figure 4 in Appendix C is suggestive of a strong relation between spouses' relative productivity and their relative weight. This evidence has already been uncovered in [Browning and Chiappori \(1998\)](#) and [Lise and Yamada \(2019\)](#).²¹ We find, in contrast with them, that the distribution of α is skewed to the right.

In our data, men work substantially more hours than their spouses despite displaying only slightly higher productivity. Equation (24) reconciles the model with the data by skewing the distribution of weights towards women, since non-work time is equivalent to leisure in our model. Implicit in our analysis, and pervasive in the literature on the taxation of couples, is the assumption that all time not spent in market activities is treated as leisure. In other words, no household production is considered. This assumption contradicts common intuition and the evidence presented by [Lise and Yamada \(2019\)](#). It is well-documented that women tend to work longer hours in home production. Therefore, it is plausible that they have lower weights even though they work fewer hours in the market, assuming all other factors are constant.

From the empirical distribution of $\boldsymbol{\iota}$, $F(\boldsymbol{\iota})$, we obtain the distribution, $\Psi(\omega)$, with the associated density, $\psi(\omega)$. We, then, compute an empirical dissonance index ξ for each family $\boldsymbol{\iota} = (w_a, w_b, \alpha)$, which works as a sufficient statistic for the dissonance in the planner's welfare metric. Relying on the fact that given our preference specification, $\xi(x, z, \boldsymbol{\iota})$ does not depend on the allocation (x, z) (see Subsection 4.1). Because $\bar{\xi}(\omega)$ is independent of (x, z) , the Pigouvian term in (23) only depends on the optimal allocation through the ratio $\mathbb{E}[x]/x(\omega)$, which simplifies our numerical procedure, as explained in Appendix C. We average it across all $\boldsymbol{\iota}$ such that $\boldsymbol{\iota} \in \mathcal{I}(\omega)$ to calculate the ω -household wedge $\bar{\xi}(\omega)$ as in equation (21). It is important to notice that $\xi(\omega)$ is averaged over families that, while displaying the same ω , may be very different in their characteristics and may reflect in a very different distribution of resources within the family and, consequently, $\xi(x, z, \boldsymbol{\iota})$. Further regularization and smoothing are needed to avoid instability on the estimates; these details

²⁰We thank an anonymous referee for suggesting using these proxies for recovering α .

²¹[Lise and Yamada \(2019\)](#) find that cross-sectional differences in expected wage profiles between a husband and wife at the time of marriage affect the allocation of private consumption expenditures and time use by households in the cross-section.

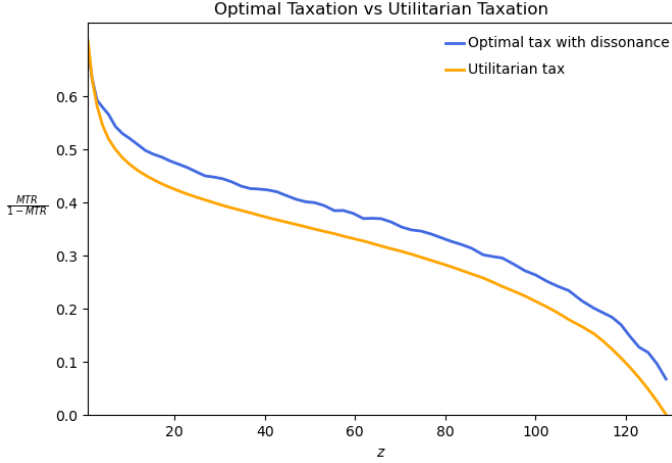


Figure 1: The solid blue line depicts the values for $T'/(1 - T')$ that satisfying equation (23) in Proposition 3 taking dissonance into account. The solid yellow line presents the household-oriented utilitarian solution ($\xi(\omega) = 1$) for the Mirrleesian problem.

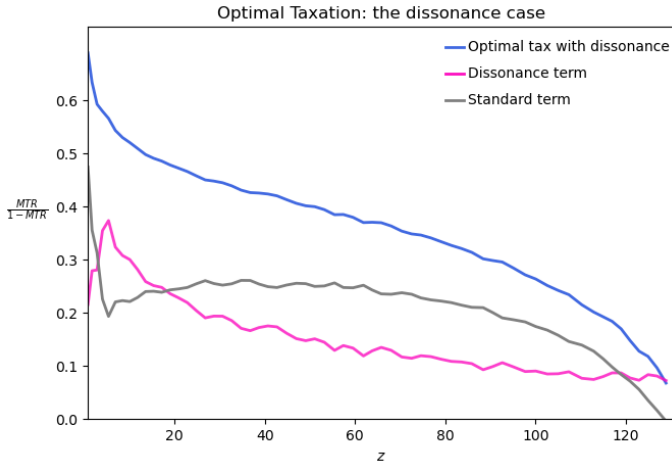


Figure 2: The solid blue line depicts the values for $T'/(1 - T')$ satisfying equation (23) in Proposition 3. The other lines decompose $T'/(1 - T')$ into a standard Mirrleesian term (gray) and a Pigouvian correction term (red) induced by dissonance.

are relegated to Appendix C.

Figure 1 displays a comparison between the optimal taxation solution derived from equation (23) in Proposition 3, which incorporates dissonance, and the household-oriented utilitarian solution (where $\xi(\omega) = 1$). The comparison reveals that dissonance leads to consistently higher marginal tax rates in this application. Corollary 1, alongside equation (24), anticipates that the planner will adjust optimal marginal tax rates, compared to the Mirrleesian benchmark, to achieve within-household redistribution. Since men have lower bargaining power, in our data, the planner aims to redistribute utility from female spouses to male spouses. According to our data, men are typically more productive, contributing more to household earnings than they consume. Thus, redistribution within these households is achieved by distorting household earnings downward through higher marginal taxes. This outcome which aligns precisely with our numerical findings illustrated in Figure 1 is, however, subject to the caveat regarding the treatment of home production previously discussed.

Figure 2 depicts the ratio of the marginal tax rate to the keep rate $T'/(1 - T')$ as a

function of the family's income z . The value of $\xi(\omega)$ is always greater than one – see Figure 5 on the Appendix C – implying a positive Pigouvian correction and an upward distortion of the marginal tax rate. The decreasing pattern of the Pigouvian correction term displayed in Figure 2 is a consequence of a relatively constant $\xi(\omega)$ and a decreasing ratio $\mathbb{E}[x]/x(\omega)$.

As mentioned previously, this joint tax system aims to balance three potentially competing objectives: efficiency, redistribution across households, and redistribution within households. The interplay between the Mirrleesian and Pigouvian terms showcases this triple trade-off. The contrast between the Mirrleesian term in Figure 2 and the Utilitarian tax in Figure 1 emphasizes that, in the presence of dissonance, the planner encounters more pronounced distortions at the household income level $z(\omega)$. These outcomes lead to amplified inefficiencies and higher redistribution across households compared to the utilitarian objective, ultimately aiming for a more significant level of intrahousehold redistribution.

6 Extensions

6.1 Generalizing Preferences and Perturbation Methods

In the preceding sections, we fully characterized optimal tax schedules under the assumption that both spouses have identical iso-elastic preferences. In this section, we use a tax perturbation approach to offer a local characterization of optimal tax schedules for collective households. This method does not depend on the Spence-Mirrlees property for preferences, enabling the inclusion of heterogeneous preferences, home production, and other aspects of household behavior. One of the main advantages of this approach is its proven effectiveness in characterizing the optimal tax rate with only a few sufficient statistics.

Proposition 4 demonstrates that, under restrictions on taxable income elasticities, the formula for optimal taxes takes a simple ABC form, as seen in [Diamond \(1998\)](#) and [Saez \(2001\)](#), with an additional term D, which we refer to as an ABC+D structure. In the general case, where there are no restrictions on elasticities, Proposition E.5 shows that the formula becomes much more complex, and additional novel sufficient statistics emerge.

This methodology involves computing the welfare and revenue effects induced by perturbing the baseline tax schedule in the direction of an arbitrary reform, denoted as $H : \mathbb{R}_+ \rightarrow \mathbb{R}$. This calculation entails identifying the Gateaux differential to accurately assess behavioral responses. The underlying principle of optimality in perturbation methods is established on the premise that there is no direction for further improvement.²²

²²[Alves et al. \(2019\)](#) show that formulae derived using variational methods may fail to characterize the

For a specific level of earnings z and within a given household ι , we introduce the conventional definitions of various types of elasticities related to taxable income, as detailed in Appendix D. These include the uncompensated elasticity, denoted by $\varepsilon(z, \iota)$, which measures the response of taxable income to changes in the net-of-tax rate without adjusting for changes in utility. The compensated elasticity, $\varepsilon^c(z, \iota)$, is analogous to the uncompensated elasticity but adjusts for these utility changes, reflecting the pure substitution effect. Additionally, we define the income elasticity, $\eta(z, \iota)$, indicating how taxable income varies with changes in non-labor income. Lastly, the cross-sectional elasticity, $\gamma(z, \iota)$, captures the variability of taxable income across different households at a given earnings level. These elasticities are crucial for understanding the behavioral responses to tax policy within our model framework.

We also use $\Phi(z)$ to represent the distribution of household income z induced by the tax system, with $\varphi(z)$ denoting the associated density. Additionally, we employ $g(z)$ to denote the average social welfare weight among all households choosing the same z . In the subsequent Proposition 4, we focus on the scenario where the elasticities are uniform among all ι -couples choosing the same z .

This constraint simplifies the expression for the optimal marginal tax rate, making it more readily comparable to our findings in Section 4 and to the formulas found in the literature that utilize unitary models for the household.

Proposition 4. *Assume, for all income levels z , that $\varepsilon(z, \iota) = \varepsilon(z)$, $\eta(z, \iota) = \eta(z)$, and $\varepsilon^c(z, \iota) = \varepsilon^c(z)$ for all families ι choosing this z . Then, in the optimal joint tax schedule, the first-order necessary condition for the marginal tax rate at a total household income of z is as follows:*

$$\frac{T'(z)}{1 - T'(z)} = A(z)B(z)C(z) + D(z), \quad (25)$$

where

$$A(z) = \frac{1}{\varepsilon^c(z)}, \quad B(z) = \frac{1 - \Phi(z)}{\varphi(z)z},$$

$$C(z) = \int_z^{\bar{z}} \{1 - g(\tilde{z})\} \exp \left\{ \int_{\tilde{z}}^{\bar{z}} \frac{\eta(\tilde{z})}{\varepsilon^c(\tilde{z})} \frac{d\tilde{z}}{\tilde{z}} \right\} \frac{\varphi(\tilde{z})}{1 - \Phi(z)} d\tilde{z},$$

and

$$D(z) = -g(z) [1 - \xi(z)].$$

optimum in models where marginal rates of substitution are not exogenously ordered by unobserved heterogeneity, a common occurrence in models with multidimensional heterogeneity.

At issue is the implicit assumption that agents are never indifferent between two different income choices, z and z' , and hence, that perturbations induce no jumps. Therefore, unless we impose more structure on agents' preferences, our model is susceptible to [Alves et al.'s \(2019\) critique](#).

Equation (25) exhibits an ABC (and D) form similar to that in [Diamond \(1998\)](#), with an additional term D accounting for the dissonance effect. See Appendix E for the proof. The term $D(z)$ serves as a Pigouvian correction, raising marginal tax rates whenever $\bar{\xi}(z) > 1$ and lowering them otherwise. Recalling that by definition

$$\xi(z, \boldsymbol{\iota}) \begin{matrix} \leq \\ \geq \end{matrix} 1 \Leftrightarrow \frac{\partial W(z - T(z), z, \boldsymbol{\iota}) / \partial z}{\partial W(z - T(z), z, \boldsymbol{\iota}) / \partial x} \begin{matrix} \leq \\ \geq \end{matrix} \frac{\partial V(z - T(z), z, \boldsymbol{\iota}) / \partial z}{\partial V(z - T(z), z, \boldsymbol{\iota}) / \partial x}.$$

Hence, marginal tax rates are reduced (increased) compared to the pure Mirrleesian optimum if, on average, a family earning z underestimates (overestimates) the disutility of effort as perceived by the planner.

In the online Appendix E, we derive Proposition 5, which presents a general formula for the optimal marginal tax rate without any restrictions on preferences. This formula incorporates welfare weights and heterogeneous elasticities at each earnings level, z . When elasticities vary across families, the optimal tax formula depends on welfare-weighted elasticities, dissonance effects, and their covariances. The extent to which preferences and household technologies can be generalized while preserving the invariance of elasticities is crucial for incorporating relevant features of household models while maintaining the tractability of unitary models.

6.2 Combining joint and individual instruments

Throughout the rest of the paper, we examine how joint tax instruments can address a significant degree of intrahousehold inequality. In this section, we move away from joint taxation by differentiating marginal tax rates for the two spouses. We introduce a low (negative) tax, denoted as $t \in \mathbb{R}$, on the gross income of the spouse a . This instrument has the potential to address the intrahousehold allocation of resources in ways that a joint tax system cannot.

Indeed, even if the planner's preferences lead to the same choices as families', the planner might still prefer a different distribution of x and z within households. This potential misalignment was inconsequential in our main analysis because the planner lacked the tools to enforce such redistribution when preferences over (x, z) were aligned. However, introducing even a simple instrument beyond income-splitting taxes enables direct influence on the internal allocation of earnings.

With the addition of t , the household's budget constraint becomes

$$x_a + x_b \leq z_a(1 - t) + z_b - T(z_a(1 - t) + z_b).$$

In a related context, Kleven et al. (2009) introduce a system where marginal tax rates incorporate some degree of dependence, moving away from an initially completely independent tax system.²³ Precisely, Kleven et al. start the analysis with individual taxes and, by employing the notion of jointness in tax schedules, investigate the feasibility of introducing dependence in marginal tax rates based on the participation of the second earner in the labor market. In contrast, our approach here starts with tax schedules characterized by income-splitting, where marginal tax rates are always equalized across spouses and introduces a separation in marginal tax rates through the parameter t .

Consider the household's optimal choice of earnings problem in the following scenario:

$$\min_{\hat{z}_a \in [0, \hat{z}]} \left\{ \alpha \left(\frac{\hat{z}_a}{\hat{w}_a} \right)^{1+\gamma} + (1 - \alpha) \left(\frac{\hat{z} - \hat{z}_a}{w_b} \right)^{1+\gamma} \right\},$$

where $\hat{z}_a \equiv z_a(1 - t)$, $\hat{z} \equiv \hat{z}_a + z_b$ and $\hat{w}_a \equiv w_a(1 - t)$. Following this minimization, we optimize with respect to $\hat{z}$. For ln preferences, the household's optimal effort allocations are given by

$$\hat{k}_a(\boldsymbol{\iota}) = \frac{\hat{w}_a^{\frac{1+\gamma}{\gamma}} (1 - \alpha)^{1/\gamma}}{\hat{w}_a^{\frac{1+\gamma}{\gamma}} (1 - \alpha)^{1/\gamma} + w_b^{\frac{1+\gamma}{\gamma}} \alpha^{1/\gamma}} \quad \text{and} \quad \hat{k}_b(\boldsymbol{\iota}) = 1 - \hat{k}_a(\boldsymbol{\iota}),$$

Here, $\hat{w}(\boldsymbol{\iota})$ denotes the adjusted household productivity

$$\hat{w}(\boldsymbol{\iota}) \equiv \left[\alpha^{\frac{-1}{\gamma}} \hat{w}_a^{\frac{1+\gamma}{\gamma}} + (1 - \alpha)^{\frac{-1}{\gamma}} w_b^{\frac{1+\gamma}{\gamma}} \right]^{\frac{\gamma}{1+\gamma}}.$$

These expressions capture the intricate dynamics of household decision-making regarding earnings under the influence of a tax t .

To understand the implications of this policy on the distribution of utility among spouses, one must simply examine

$$V_a - V_b = \ln \frac{\alpha}{1 - \alpha} - \left[\left(\frac{\hat{k}_a}{\hat{w}_a} \right)^{1+\gamma} - \left(\frac{\hat{k}_b}{w_b} \right)^{1+\gamma} \right] \frac{\hat{z}^{1+\gamma}}{1 + \gamma}. \quad (26)$$

It is immediate to see that for a fixed α , the reform cannot influence the relative consumption of spouses. Therefore, the impact of introducing a subsidy ($t < 0$) on a 's earn-

²³They also calculate the optimum by simplifying the multidimensional nature of the problem. They assume that primary earners make choices along the intensive margin, while secondary earners, who have identical labor market productivities, make choices along the extensive margin only. Heterogeneity among secondary earners is considered in terms of their utility cost of working and/or benefit of home production.

ings depends solely on how it affects the second term on the right-hand side of (26). The first consequence of this small subsidy is to induce a variation in z by increasing $\hat{w}(\iota)$. Assuming $V_a < V_b$, this reduces the utility difference between the husband and the wife if $l_a < l_b$ and increases it otherwise. Of course, this could have been accomplished more efficiently by simply changing $T(\cdot)$. The novel consequence arises through the impact on $(\hat{k}_a/\hat{w}_a)^{1+\gamma} - (\hat{k}_b/w_b)^{1+\gamma}$. For a large set of parameter values, subsidizing the secondary earner while using $T(\cdot)$ to keep (x, z) fixed will generate a counter-intuitive and non-intended result of reducing a 's utility if and only if $k_a < (1 + \gamma)^{-1}$ — see the online Appendix F.²⁴

Policy-dependent α . The counter-intuitive result that subsidizing a spouse's earnings reduces her utility relies on the assumption that the reform does not impact α . Immervoll et al. (2011), who also make this independence assumption regarding the Pareto weights, justify it by assuming that α depends on the utilities attained by singles, which, according to their claim, remain unaffected by the taxation of couples. It is essential to note, however, that if power is determined by the 'shadow of marriage markets', as suggested by Becker (1973, 1974), we need to explicitly address how marriage markets are affected by the reform to make such statements. Rather than pursuing this approach, we provide an example to illustrate how the results may be overturned if we allow the policy to affect spouses' power within the collective framework

The example is based on an additional potential impact of subsidizing a : to increase her bargaining power by enhancing her (net) relative earnings potential. In our baseline implementation, we extracted α s from the data and demonstrated how they increase with w_a/w_b .

For the sake of simplicity, let's assume that this relationship is deterministic and given by $\hat{\alpha} = \hat{w}_a/(\hat{w}_a + w_b)$, and the term in brackets in (26) does not depend on t . The only consequence of subsidizing a 's earnings is to transfer consumption from b to a , restoring the intuitive result.

This discussion's purpose is not to dismiss the counter-intuitive result but to highlight that the framework underlying most findings in optimal taxation of couples may conflict with some gender-based policies advocated and adopted in many parts of the world; the latter cannot be rationalized by a fixed α approach.

²⁴This condition can be written solely in terms of primitives as $\gamma^\gamma(1 - \alpha)w_a^{1+\gamma} < \alpha w_b^{1+\gamma}$.

7 Conclusion

In this paper, we derive optimal tax schedules for multiperson households, under the constraint that taxes involve income-splitting. Leveraging a commonly adopted specification for preferences enables us to address the multidimensional screening problem inherent in optimal household taxation and to compute the optimal tax schedule. Our findings underscore the quantitative impact of dissonance, which reflects the misalignment between the household's and planner's objectives regarding household income taxes— a tool that can only indirectly influence intrahousehold inequality. Notably, optimal marginal tax rates may deviate from non-negativity, contrasting with a well-established property derived for single-person households in the Mirrlees tradition.

Moreover, we broaden our analysis to investigate the implications of introducing differentiation in the marginal tax rates faced by the two spouses. Our findings suggest that it is typically optimal to impose a minor tax on the spouse that society seeks to promote, as income taxes effectively subsidize the spouse's leisure. However, the validity of this conclusion hinges crucially on the assumption that power remains unaffected by choices or the tax system (or both). It is evident that if earnings, a policy-dependent variable, rather than innate productivity, a policy-independent variable, determine power, these results may be reversed.

While our model is exclusively focused on labor market effort, it overlooks the substantial amount of time that agents, often those with less power within the household, dedicate to home production. This aspect is not captured within our current framework. Including home production would disrupt the aggregation result convenient for applying a mechanism design approach, justifying its exclusion from this paper. However, it is worth contemplating the potential implications that incorporating home production could have.

An increase in marginal tax rates is likely to incentivize spouses to engage more in home production, with the secondary earner possibly devoting a larger fraction of their working time to such activities. The distribution of increased home production between spouses would depend on various factors, such as their initial involvement in home production. While a higher marginal tax rate may elevate home production and the relative effort of the secondary earner, further exploration of these dynamics, potentially through parametric solutions to the tax problem, presents an intriguing avenue for future research.

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A Proofs

Proof of Proposition 1. Recall that

$$W(x, z, \boldsymbol{\iota}) = \frac{1}{2} \left[u_a(x_a(z - T(z))) - h_a \left(\frac{z_a(z)}{w_a} \right) \right. \\ \left. + u_b(z - T(z) - x_a(z - T(z))) - h_b \left(\frac{z - z_a(z)}{w_b} \right) \right],$$

which implies

$$\frac{dW(x, z, \boldsymbol{\iota})}{dz} = \frac{1}{2} \left[u'_a(x_a) [1 - T'(z)] x'_a(x) - h'_a \left(\frac{z_a}{w_a} \right) \frac{z'_a(z)}{w_a} \right. \\ \left. + u'_b(x_b) [1 - x'_a(x)] [1 - T'(z)] - h'_b \left(\frac{z_b}{w_b} \right) \frac{1 - z'_a(z)}{w_b} \right].$$

Using the first order conditions with respect to x_a and h_a ,

$$\alpha u'_a(x_a) = (1 - \alpha) u'_b(x - x_a),$$

and,

$$\alpha h'_a(z_a/w_a)/w_a = (1 - \alpha) h'_b((z - z_a)/w_b)/w_b,$$

respectively, we get

$$\frac{dW(x, z, \boldsymbol{\iota})}{dz} = \frac{1}{2[1 - \alpha]} u'_a(x_a) \left[[1 - T'(z)] \{ [1 - 2\alpha] x'_a(x) + \alpha \} \right. \\ \left. - h'_a \left(\frac{z_a}{w_a} \right) \frac{1}{u'_a(x_a) w_a} [[1 - 2\alpha] z'_a(z) + \alpha] \right],$$

or

$$\frac{dW(x, z, \boldsymbol{\iota})}{dz} = \frac{1}{2[1 - \alpha]} u'_a(x_a) [1 - T'(z)] \left\{ \{ [1 - 2\alpha] x'_a(x) + \alpha \} \right. \\ \left. - \frac{h'_a(z_a/w_a)}{[1 - T'(z)] u'_a(x_a) w_a} [[1 - 2\alpha] z'_a(z) + \alpha] \right\},$$

and, finally,

$$\frac{dW(x, z, \boldsymbol{\iota})}{dz} = \frac{u'_a(x_a) [1 - T'(z)]}{2[1 - \alpha]} \{[1 - 2\alpha] [x'_a(x) - z'_a(z)]\}.$$

For the last part, taking the total derivative of W with respect to z , using the definitions of $\mathcal{U}(x, \boldsymbol{\iota})$, $\mathcal{V}(x, \boldsymbol{\iota})$, $\mathcal{H}(z, \boldsymbol{\iota})$ and $\mathcal{K}(z, \boldsymbol{\iota})$, we have

$$\begin{aligned} \frac{dW(x, z, \boldsymbol{\iota})}{dz} &= \frac{\partial \mathcal{V}(x, \boldsymbol{\iota})}{\partial x} [1 - T'(z)] - \frac{\partial \mathcal{K}(z, \boldsymbol{\iota})}{\partial z} = \\ &\left[\frac{\partial \mathcal{V}(x, \boldsymbol{\iota})}{\partial x} \left(\frac{\partial \mathcal{U}(x, \boldsymbol{\iota})}{\partial x} \right)^{-1} - \frac{\partial \mathcal{K}(z, \boldsymbol{\iota})}{\partial z} \left(\frac{\partial \mathcal{H}(z, \boldsymbol{\iota})}{\partial z} \right)^{-1} \right] \frac{\partial \mathcal{H}(z, \boldsymbol{\iota})}{\partial z}. \quad (\text{A.1}) \end{aligned}$$

Using the definition of $\xi(x, z, \boldsymbol{\iota})$, the result follows. □

Lemma 2. *If we define*

$$\sigma_i(x_i) \equiv \frac{-u''_i(x_i)x_i}{u'_i(x_i)} \quad \text{and} \quad \gamma_i(l_i) \equiv \frac{h''_i(l_i)l_i}{h'_i(l_i)},$$

then

$$x'_a(x) = \frac{x_a}{\sigma_a(x_a)} \left[\frac{x_b}{\sigma_b(x_b)} + \frac{x_a}{\sigma_a(x_a)} \right]^{-1} \quad \text{and} \quad z'_a(z) = \frac{z_a}{\gamma_a(l_a)} \left[\frac{z_b}{\gamma_b(l_b)} + \frac{z_a}{\gamma_a(l_a)} \right]^{-1}.$$

Proof. The consumption-sharing program is

$$\max_{x_a} \alpha u_a(x_a) + [1 - \alpha] u_b(x - x_a).$$

The first-order condition is

$$\alpha u'_a(x_a) - [1 - \alpha] u'_b(x - x_a) = 0. \quad (\text{A.2})$$

Taking the total derivative

$$[\alpha u''_a(x_a) + [1 - \alpha] u''_b(x - x_a)] dx_a - [1 - \alpha] u''_b(x - x_a) dx = 0,$$

which rearranging and using (A.2) leads to the Borch rule.

The family's cost-minimization problem is

$$\min_{z_a} \left\{ \alpha h_a \left(\frac{z_a}{w_a} \right) + (1 - \alpha) h_b \left(\frac{z - z_a}{w_b} \right) \right\},$$

which has as a necessary and sufficient condition for a solution:

$$\alpha h'_a \left(\frac{z_a}{w_a} \right) \frac{1}{w_a} - (1 - \alpha) h'_b \left(\frac{z - z_a}{w_b} \right) \frac{1}{w_b} = 0. \quad (\text{A.3})$$

Differentiating (A.3), we get

$$\left[\alpha h''_a(l_a) \frac{1}{w_a^2} + (1 - \alpha) h''_b(l_b) \frac{1}{w_b^2} \right] dz_a = (1 - \alpha) h''_b(l_b) \frac{1}{w_b^2} dz.$$

Dividing both sides by $(1 - \alpha) h'_b(l_b) \frac{1}{w_b}$ and using (A.3), we have

$$\left[\frac{h''_a(l_a)}{h'_a(l_a)} \frac{1}{w_a} + \frac{h''_b(l_b)}{h'_b(l_b)} \frac{1}{w_b} \right] dz_a = \frac{h''_b(l_b)}{h'_b(l_b)} \frac{1}{w_b} dz,$$

which we can further manipulate by multiplying by z_b

$$\left[\frac{h''_a(l_a)}{h'_a(l_a)} \frac{z_b}{w_a} + \frac{h''_b(l_b)}{h'_b(l_b)} \frac{z_b}{w_b} \right] dz_a = \frac{h''_b(l_b)}{h'_b(l_b)} \frac{z_b}{w_b} dz$$

and reorganizing

$$\left[\frac{h''_a(l_a)}{h'_a(l_a)} n_a \frac{z_b}{z_a} + \frac{h''_b(l_b)}{h'_b(l_b)} l_b \right] dz_a = \frac{h''_b(l_b)}{h'_b(l_b)} l_b dz$$

to obtain

$$dz_a = \frac{\gamma_b(l_b)/z_b}{\gamma_a(l_a)\frac{1}{z_a} + \gamma_b(l_b)\frac{1}{z_b}} dz,$$

or

$$dz_a = \frac{\gamma_b(l_b)/l_b}{\frac{w_b}{w_a} \gamma_a(l_a)/l_a + \gamma_b(l_b)/l_b} dz,$$

which we further manipulate to obtain

$$dz_a = \frac{z_a/\gamma_a(l_a)}{z_b/\gamma_b(l_b) + z_a/\gamma_a(l_a)} dz.$$

Naturally, $dz_b/dz = 1 - dz_a/dz$.

□

Proof of Proposition 2. Inserting in (A.1) the expressions for $x'_a(x)$ and $z'_a(z)$ from Lemma 2 yields the result.

□

Proof of Lemma 1. To formulate the planner's problem, recall that when agents' utilities are ln-isoelastic as in (14) then $\partial \mathcal{U}(x, \iota)/\partial x = x^{-1} = \partial \mathcal{V}(x, \iota)/\partial x$. Moreover, we can write $\mathcal{K}(z, \iota) = \mathcal{A}_K(\iota)z^{1+\gamma}$ and $\mathcal{H}(z, \iota) = \mathcal{A}_H(\iota)z^{1+\gamma}$.²⁵ In this case, the dissonance term is independent of (x, z) ,

$$\xi(x, z, \iota) = \frac{\partial \mathcal{K}(z, \iota)/\partial z}{\partial \mathcal{H}(z, \iota)/\partial z} = \frac{\mathcal{A}_K(\iota)}{\mathcal{A}_H(\iota)} \equiv \hat{\xi}(\iota),$$

and, conveniently,

$$\begin{aligned} \mathcal{H}(z, \iota) - \mathcal{K}(z, \iota) &= [\mathcal{A}_H(\iota) - \mathcal{A}_K(\iota)] z^{1+\gamma} = \left[1 - \frac{\mathcal{A}_K(\iota)}{\mathcal{A}_H(\iota)}\right] \mathcal{A}_H(\iota) z^{1+\gamma} \\ &= [1 - \hat{\xi}(\iota)] \mathcal{H}(z, \iota). \end{aligned}$$

Since

$$\mathcal{H}(z, \iota) = \frac{1}{1+\gamma} \left(\frac{z}{\omega}\right)^{1+\gamma}, \quad \forall \iota \in \mathcal{I}(\omega),$$

then

$$\mathbb{E} [\mathcal{H}(z, \iota)[1 - \xi(x, z, \iota)] | \iota \in \mathcal{I}(\omega)] = \frac{1}{1+\gamma} \left(\frac{z}{\omega}\right)^{1+\gamma} \mathbb{E} [1 - \xi(x, z, \iota) | \iota \in \mathcal{I}(\omega)]$$

□

Proof of Proposition 3. The planner's program is

$$\max_{(x(\omega), z(\omega))_\omega} \int_{\underline{\omega}}^{\bar{\omega}} \left\{ \bar{V}(x(\omega), z(\omega), \omega) + [1 - \bar{\xi}(\omega)] \frac{1}{1+\gamma} \left(\frac{z(\omega)}{\omega}\right)^{1+\gamma} \right\} \psi(\omega) d\omega, \quad (\text{A.4})$$

subject to

$$\omega \in \operatorname{argmax}_{\hat{\omega}} V(x(\hat{\omega}), z(\hat{\omega}), \omega),$$

and

$$\int_{\underline{\omega}}^{\bar{\omega}} [z(\omega) - x(\omega)] \psi(\omega) d\omega \geq G. \quad (\text{A.5})$$

For all ω , let $v(\omega) = \max_{\hat{\omega}} V(x(\hat{\omega}), z(\hat{\omega}), \omega)$ and define $x(v(\omega), z(\omega), \omega)$ by

$$v(\omega) \equiv V(x(v(\omega), z(\omega), \omega), z(\omega), \omega).$$

²⁵ $\mathcal{A}_H(\iota) = \omega(\iota)^{-1-\gamma}/(1+\gamma)$ as shown in (17). As for $\mathcal{A}_K(\iota)$, its expression can be derived by plugging k_a and k_b in (16) in the general formulas for $\mathcal{K}(z, \iota)$.

Then, the incentive compatibility constraint is equivalent to

$$\dot{v}(\omega) = \frac{\partial}{\partial \omega} V(x(\omega), z(\omega), \omega), \quad (\text{A.6})$$

and $z(\omega)$ non-increasing in ω .

Ignoring, as usual, the monotonicity constraint, the planner's program is to maximize (A.4) subject to (A.5) and (A.6). The Hamiltonian associated with this problem is

$$\begin{aligned} \mathcal{H}(v(\omega), z(\omega), \mu(\omega), \lambda, \omega) = & \left\{ v(\omega) + [1 - \bar{\xi}(\omega)] \frac{1}{1 + \gamma} \left(\frac{z(\omega)}{\omega} \right)^{1+\gamma} + \right. \\ & \left. \lambda [z(\omega) - x(v(\omega), z(\omega), \omega)] \right\} \psi(\omega) - \mu(\omega) \frac{z(\omega)^{1+\gamma}}{\omega^{2+\gamma}}, \end{aligned}$$

where $\lambda \in \mathbb{R}_+$ is the Lagrange multiplier associated with the government budget constraint and $\mu(\omega)$ is the co-state variable. At the optimum, the control $z(\omega)$ maximizes the Hamiltonian function giving the following first-order condition:

$$[1 - \bar{\xi}(\omega)] \frac{z(\omega)^\gamma}{\omega^{1+\gamma}} \psi(\omega) + \lambda \left[1 - \frac{\partial x(v(\omega), z(\omega), \omega)}{\partial z} \right] \psi(\omega) = -\mu(\omega) \frac{z(\omega)^\gamma}{\omega^{2+\gamma}}.$$

Noting that

$$\frac{\partial x(v(\omega), z(\omega), \omega)}{\partial z} = 1 - T'(z(\omega))$$

and $z(\omega)^\gamma / \omega^{1+\gamma} = (1 - T'(z(\omega))) / x(\omega)$, we get

$$[1 - \bar{\xi}(\omega)] [1 - T'(z(\omega))] \psi(\omega) + \lambda T'(z(\omega)) x(\omega) \psi(\omega) = -\mu(\omega) [1 - T'(z(\omega))] / \omega,$$

which can be re-arranged to read

$$\frac{T'(z(\omega))}{1 - T'(z(\omega))} = \frac{1}{\lambda x(\omega) \psi(\omega) \omega} \mu(\omega) - \frac{1}{\lambda x(\omega)} [1 - \bar{\xi}(\omega)].$$

As for $\mu(\omega)$, the other first-order condition of the Hamiltonian is

$$\dot{\mu}(\omega) = -\frac{\partial}{\partial v} \mathcal{H}(\cdot) = \left[\lambda \frac{\partial x(v(\omega), z(\omega), \omega)}{\partial v} - 1 \right] \psi(\omega)$$

and the transversality conditions $\mu(\underline{\omega}) = \mu(\bar{\omega}) = 0$. Integrating on both sides from $\underline{\omega}$ to ω and applying the fundamental theorem of calculus we have

$$\mu(\omega) - \mu(\underline{\omega}) = \int_{\underline{\omega}}^{\omega} \dot{\mu}(s) ds = \int_{\underline{\omega}}^{\omega} \left[\lambda \frac{\partial x(v(s), z(s), s)}{\partial v} - 1 \right] \psi(s) ds.$$

Using the transversality condition we have

$$\mu(\omega) = \lambda \int_{\underline{\omega}}^{\omega} \left[\frac{\partial x(v(s), z(s), s)}{\partial v} - \lambda^{-1} \right] \psi(s) ds. \quad (\text{A.7})$$

Note that from the transversality condition $\mu(\underline{\omega}) = 0$ we can pin down the social value of an extra unit of income:

$$0 = \mu(\underline{\omega}) = \lambda \int_{\underline{\omega}}^{\bar{\omega}} \left[\frac{1}{\lambda} - \frac{\partial x(v(\omega), z(\omega), \omega)}{\partial v} \right] \psi(\omega) d\omega.$$

Therefore,

$$\lambda = \left\{ \int_{\underline{\omega}}^{\bar{\omega}} \frac{\partial x(v(\omega), z(\omega), \omega)}{\partial v} \psi(\omega) d\omega \right\}^{-1},$$

or $\lambda = \mathbb{E}[x]^{-1}$, since $\partial x(v(\omega), z(\omega), \omega)/\partial v = x(v(\omega), z(\omega), \omega)$, for all ω . Using all to produce the final formula

$$\frac{T'(z(\omega))}{1 - T'(z(\omega))} = \frac{1 + \gamma}{x(\omega)\psi(\omega)\omega} \int_{\underline{\omega}}^{\omega} [\mathbb{E}[x] - x(s)] \psi(s) ds - \frac{\mathbb{E}[x]}{x(\omega)} [1 - \bar{\xi}(\omega)].$$

□

B Aggregation with $\sigma \neq 1$

In Section 4 we obtain the aggregation result for preferences that take the log-isoelastic particular form. In what follows, we generalize this result. Assume that preferences are identical and iso-elastic, i.e., for $i = a, b$

$$U_i(x_i, l_i) = \frac{x_i^{1-\sigma} - 1}{1 - \sigma} - \frac{l_i^{1+\gamma}}{1 + \gamma},$$

where $\sigma \in (0, 1]$ and $\gamma > 0$ (let $u(x_i) = \ln x_i$ for $\sigma = 1$). Applying Borch's rule to determine the allocation of consumption in this case yields the following

$$\mathcal{U}(x, \boldsymbol{\iota}) = \max_{x_a} \left\{ \alpha \frac{x_a^{1-\sigma} - 1}{1 - \sigma} + (1 - \alpha) \frac{(x - x_a)^{1-\sigma} - 1}{1 - \sigma} \right\} = \frac{B(\alpha)x^{1-\sigma} - 1}{1 - \sigma},$$

where $B(\alpha) = [\alpha^{1/\sigma} + (1 - \alpha)^{1/\sigma}]^\sigma$. Combining this expression with the one we found for $\mathcal{H}(z, \boldsymbol{\iota})$ we have:

$$V(x, z, \boldsymbol{\iota}) = \mathcal{U}(x, \boldsymbol{\iota}) - \mathcal{H}(z, \boldsymbol{\iota}) = B(\alpha) \left[\frac{x^{1-\sigma} - B(\alpha)^{-1}}{1 - \sigma} - \theta(\boldsymbol{\iota}) \frac{z^{1+\gamma}}{1 + \gamma} \right],$$

where

$$\theta(\iota) \equiv \frac{\omega(\iota)^{-1-\gamma}}{[\alpha^{1/\sigma} + (1-\alpha)^{1/\sigma}]^\sigma}.$$

Household preferences are represented by

$$u(x) - \theta(\iota)h(z) \tag{B.1}$$

where $\theta(\iota)$ now defines the household type.

We focus on the case $\sigma = 1$ because it facilitates the implementation of our program. The counterparts for the dissonance expressions we derive are a straightforward generalization of the ones used here. Specializing for ln preferences eliminates only one feature, which is the possible dependence of marginal social welfare weights on α . In fact, the marginal social value of one unit of consumption in the hands of a couple with net income x and household weight α is $0.5 [(\alpha)^{1/\sigma} + (1-\alpha)^{1/\sigma}]^\sigma x^{1-\sigma}/(1-\sigma)$ for $\sigma \neq 1$ and $\ln x$ for $\sigma = 1$. While interesting *per se*, the possibility that Pareto weights vary with aspects of an agent type has already been studied in other contexts, e.g., [Choné and Laroque \(2010\)](#), so we skip it for brevity.

C Implementation Algorithm for the Numeric Simulations

In this section, we describe our numerical approach to compute the optimal tax schedules derived in Proposition 3. We utilize real data from the 2017 March CPS to gain insights into the shape of optimal taxation and the impact of dissonance. Specifically, we focus on couples without children, employing hourly wages as a proxy for individual productivity and hours worked per week to measure labor supply.

Calculate α : For each (w_a, w_b) we generate α by solving equation (24) for α , which gives

$$\alpha = \frac{w_a l_b^\gamma}{w_a l_b^\gamma + w_b l_a^\gamma}. \tag{C.1}$$

Calculate ξ : For each family ι calculate $\xi(\iota)$ as

$$\xi(\iota) = \frac{\frac{1}{2} \left(\frac{k_a}{w_a} \right)^{1+\gamma} + \frac{1}{2} \left(\frac{k_b}{w_b} \right)^{1+\gamma}}{\alpha \left(\frac{k_a}{w_a} \right)^{1+\gamma} + (1-\alpha) \left(\frac{k_b}{w_b} \right)^{1+\gamma}}, \tag{C.2}$$

where k_i for $i = a, b$ is defined in equation (16) and we relied on ξ 's being independent of

(x, z) .

This procedure gives us the primitive distribution $F(\iota)$ from which we may define $\Psi(\omega)$ by $\Psi(\omega) = F(\iota | \iota \in \mathcal{G}(\omega'); \omega' \leq \omega)$.

After calculating these objects, we simulate the marginal tax rate by adapting the algorithm proposed by [Mankiw et al. \(2009\)](#), based on a discrete grid of the empirical distributions. With some abuse, let $\psi(\omega)$ denote the associated density. For every ω we also calculate $\bar{\xi}(\omega)$ from its definition, (21).

Step 1: Given $\psi(\omega)$, discretize the probability mass function $\pi(\omega)$ in the following way: divide the interval considered $[\underline{\omega}, \bar{\omega}]$ in N bins of equal bandwidth Δ , creating a grid where $\{\omega_i\}_{i=1}^N$ with $\underline{\omega} \leq \omega_1 \leq \dots \leq \omega_n \leq \dots \leq \omega_N \leq \bar{\omega}$ are the midpoints in the grid. The probability mass function at each ω is given by

$$\pi(\omega) = \Psi(\omega + \Delta/2) - \Psi(\omega - \Delta/2),$$

where $\Psi(\cdot)$ is the cumulative distribution function of ω .

Step 2: For each $\{\omega_i\}_{i=1}^N$ calculate the discretized $\bar{\xi}(\omega_i)$ through

$$\bar{\xi}(\omega_i) = \frac{1}{\#\mathcal{G}(\omega_i)} \sum_{\iota \in \mathcal{G}(\omega_i)} \xi(\iota),$$

to smooth out the function ξ we apply a moving average procedure on it.

Now we start the loop for the calculation of the optimal marginal tax rate $T_k(\omega_i)$.

Step 3: Start with a simple guess for the marginal tax rate. For instance, a flat tax schedule $T_0(\omega_i) = .35$ for all ω_i in the grid and a lump-sum transfer $t_0 = .0001$.

Step 4: Since $T_{k-1}(\cdot)$ at each iteration is only calculated at points of the grid, extrapolate $T_{k-1}(\cdot)$ to be defined as well at points outside the grid in the following way:

$$T'_{k-1}(\omega) = \begin{cases} T'_{k-1}(\omega_1), & \text{for } \underline{\omega} \leq \omega \leq \omega_1; \\ T'_{k-1}(\omega_2), & \text{for } \omega_1 \leq \omega \leq \omega_2; \\ \dots & \\ T'_{k-1}(\omega_n), & \text{for } \omega_{n-1} \leq \omega \leq \omega_n; \\ \dots & \\ T'_{k-1}(\omega_N), & \text{for } \omega_{N-1} \leq \omega \leq \bar{\omega}; \end{cases}$$

generating the tax schedule

$$T_{k-1}(\omega_i) = \begin{cases} T'_{k-1}(\omega_1)z(\omega_i) - t_{k-1}, & \text{for } \omega \leq \omega_i \leq \omega_1; \\ T'_{k-1}(\omega_1)z(\omega_1) + T'_{k-1}(\omega_2)[z(\omega_i) - z(\omega_1)] - t_{k-1}, & \text{for } \omega_1 \leq \omega_i \leq \omega_2; \\ \dots \\ T'_{k-1}(\omega_1)z(\omega_1) + \sum_{j=2}^{n-1} T'_{k-1}(\omega_j)[z(\omega_j) - z(\omega_{j-1})] + \\ T'_{k-1}(\omega_n)[z(\omega_i) - z(\omega_{n-1})] - t_{k-1}, & \text{for } \omega_{n-1} \leq \omega_i \leq \omega_n; \end{cases}$$

Step 5: Given $T_{k-1}(\cdot)$, calculate the optimal labor supply at each ω_i by solving

$$\max_z \ln(z - T_{k-1}(z)) - \frac{(z/\omega_i)^{1+\gamma}}{1+\gamma}.$$

The necessary first-order condition for the type ω_i problem is

$$1 - T'_{k-1}(z) = \omega_i^{-(1+\gamma)} z^\gamma (z - T_{k-1}(z)) \quad (\text{C.3})$$

and the optimal $z^*(\omega_i)$ is implicitly defined in this first-order condition.

Step 6: Given the optimal choice of $z(\omega_i)$, we can define the consumption as

$$x(\omega_i) = z^*(\omega_i) - T_{k-1}(z^*(\omega_i))$$

and use all the information to update the marginal tax rate at each point of the grid using

$$\frac{T'_k(\omega_i)}{1 - T'_k(\omega_i)} = \frac{1 + \gamma}{x(\omega_i)\psi(\omega_i)\omega_i} \int_{\omega_i}^{\bar{\omega}} [x(s) - \mathbb{E}^\psi[x]] \psi(s) ds - \frac{\mathbb{E}^\psi[x]}{x(\omega_i)} [1 - \bar{\xi}(\omega_i)]. \quad (\text{C.4})$$

Since the family utility satisfies the single-crossing property w.r.t. ω_i , this first-order condition and the monotonicity of $z^*(\omega_i)$ are sufficient for optimality.

Step 7: The formula in *Step 6* is unfeasible to estimate given our approximation of the wage distribution in the grid. Therefore, we must use a discrete approximation. One possibility is as follows:

$$\frac{T'_k(\omega_i)}{1 - T'_k(\omega_i)} \approx \frac{1 + \gamma}{x(\omega_i)\omega_i} \frac{1}{\pi(\omega_i)/\Delta} \sum_{j=i+1}^N [x(\omega_j) - \hat{\mathbb{E}}[x]] \pi(\omega_j) - \frac{\hat{\mathbb{E}}[x]}{x(\omega_i)} [1 - \bar{\xi}(\omega_i)]. \quad (\text{C.5})$$

where $\bar{\xi}(\omega_i)$ is calculated at *Step 2* and

$$\hat{\mathbb{E}}[x] = \sum_{j=1}^N x(\omega_j) \pi(\omega_j).$$

Step 8: Update t to ensure the budget constraint in the following way:

$$t_k = \sum_{j=1}^N T_k(w_j) \pi(\omega_j).$$

Step 9: Return to *Step 4* until

$$\max_i \{|T'_k(\omega_i) - T'_{k-1}(\omega_i)|\} < 10^{-3}.$$

Step 10: After getting a fixed point of this operator, the only thing to be checked is the monotonicity to guarantee that the found tax schedule is implementable.

Note that this approximation is increasingly precise as $\Delta \rightarrow 0$ and $\omega_N \rightarrow +\infty$.

C.1 Figures from the Numerical Exercise

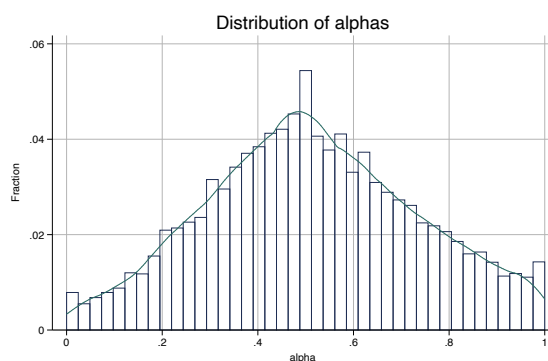


Figure 3: This figure presents the histogram of raw alphas calculated according to equation (24).

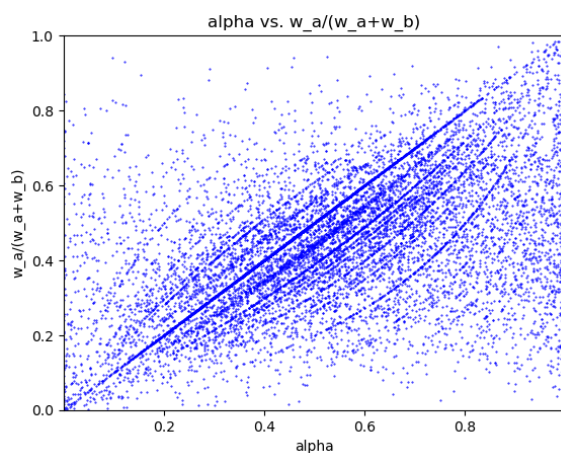


Figure 4: This figure illustrates the relation between weights α and productivity ratio uncovered by [Browning and Chiappori \(1998\)](#) and [Lise and Yamada \(2019\)](#) is also presented in our data when weights are implied by equation (24).

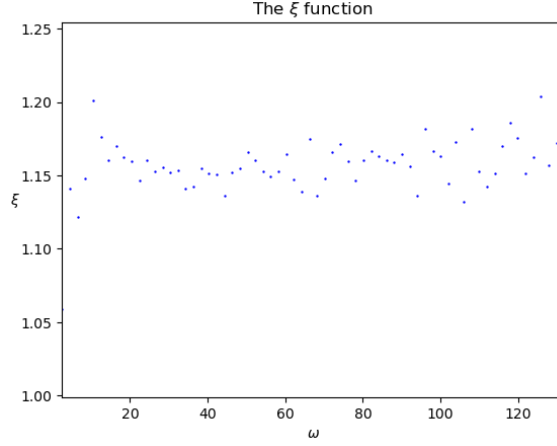


Figure 5: This figure presents the empirical function $\xi(w)$ implied from the data and equation (24).

D Elasticities

We first recall that in [Saez's](#) original paper, elasticities were defined for linearized budget sets. Here, we follow, instead, [Jacquet et al. \(2013\)](#) and [Scheuer and Werning \(2017\)](#) in using the elasticities defined under nonlinear budget sets. That is, we define the elasticity of taxable income with respect to the retention rate $1 - T'(z)$, $\epsilon(z, \iota)$, as

$$\epsilon(z, \iota) = \Delta^{-1} \left[\partial_{xz} V(x, z, \iota) z - \partial_{xx} V(x, z, \iota) \frac{\partial_z V(x, z, \iota)}{\partial_x V(x, z, \iota)} z - \partial_x V(x, z, \iota) \right] \frac{1 - T'(z)}{z},$$

where

$$\Delta = \left[\partial_{xx} V(x, z, \iota) \left(\frac{\partial_z V(x, z, \iota)}{\partial_x V(x, z, \iota)} \right)^2 - 2 \partial_{xz} V(x, z, \iota) \frac{\partial_z V(x, z, \iota)}{\partial_x V(x, z, \iota)} + \partial_{zz} V(x, z, \iota) - \partial_x V(x, z, \iota) T''(z) \right].$$

The last term in Δ , $\partial_x V(x, z) T''(z)$, captures the curvature in the budget set. [Saez \(2001\)](#), in contrast, defines the elasticity using $\tilde{\Delta} = \Delta + \partial_x V(x, z) T''(z)$. The advantage of using $\tilde{\Delta}$ is its familiarity, whereas the advantage of using Δ is the simplification of optimal tax formulae.

We can also define the income elasticity of taxable earnings

$$\eta(z, \iota) = \Delta^{-1} \left[\partial_{xx} V(x, z, \iota) \frac{-\partial_z V(x, z, \iota)}{\partial_x V(x, z, \iota)} + \partial_{xz} V(x, z, \iota) \right] (1 - T'(z))$$

and, using the Slutsky equation, we can compute the compensated elasticity

$$\epsilon^c(z, \boldsymbol{\iota}) = \epsilon(z, \boldsymbol{\iota}) - \eta(z, \boldsymbol{\iota}) = \Delta^{-1} [\partial_x V(x, z, \boldsymbol{\iota})] \frac{1 - T'(z)}{z}.$$

Online Appendix – not for publication

E Generalizing Preferences: The Tax Perturbation Approach

In the previous sections, we were able to fully characterize the optimal tax schedules when spouses share the same iso-elastic preferences. This section uses a tax perturbation approach to provide expressions for the optimal marginal tax rate for the case in which preferences are fully general.

The first step is to compute the welfare and revenue effects of perturbing the baseline schedule in the direction of an arbitrary reform $H : \mathbb{R}_+ \rightarrow \mathbb{R}$, which amounts to computing its Gateaux differential. That is, consider a reform whereby the current schedule is replaced by a new one, $\tilde{T} = T + \mu H$, for $\mu \in \mathbb{R}$. When the reform H is a C^2 function, for small values of μ the resulting tax schedule is a *perturbed* version of the baseline tax and lies within its neighborhood. This section restricts taxes to the class of joint taxation $\mathcal{T}_0$. Given a baseline tax system, $T \in \mathcal{T}_0$, we say that the perturbation $H : \mathbb{R}_+ \rightarrow \mathbb{R}$ is *within the same class* if the perturbed tax schedule remains in the same class as T , that is, $\tilde{T} = T + \mu H \in \mathcal{T}_0$ for small enough μ . We will then say that the tax schedule is optimal within this class if there is no perturbation *within the same class*, which improves the planner's objective.^{26,27}

We start by deriving the behavioral response for the change in a baseline tax schedule $T : \mathbb{R}_+ \rightarrow \mathbb{R}$ (a candidate for the optimal schedule) in the direction of the reform $H : \mathbb{R}_+ \rightarrow \mathbb{R}$. Consider the program of a type ι household facing a nonlinear tax schedule $T : \mathbb{R}_+ \rightarrow \mathbb{R}$:

$$\max_{(x,z) \in \mathbb{R}_+^2} \{V(x, z, \iota) \quad \text{s.t.} \quad x = z - T(z)\}. \quad (\text{E.1})$$

The solution to this program defines the earnings supply functional $z_\iota : \mathcal{T} \rightarrow \mathbb{R}$, where $\mathcal{T}$ is the class of tax schedules being considered. We measure *behavioral responses* to the reform in the direction of H by the Gateaux derivative of the earnings supply functional, denoted by $dz_\iota(T; H)$ (see (E.7) for the formal definition).

²⁶There may still be H' for which $\tilde{T}' = T + \mu H' \notin \mathcal{T}_0$, which improves the planner's objective. In fact, in Subsection 6.2 we study reforms whose resulting perturbed classes are not within the class of joint taxation $\mathcal{T}_0$.

²⁷Alves et al. (2019) show that formulae derived using variational methods may fail to characterize the optimum in models in which marginal rates of substitution are not exogenously ordered according to the unobserved heterogeneity, which typically arises in models with multidimensional heterogeneity. At issue is the implicit assumption that agents are never indifferent between two choices z and z' with $z > z'$ and, hence, that perturbations induce no jumps. Therefore, unless we impose more structure on agents' preferences, our model is susceptible to Alves et al.'s (2019) critique.

The overall marginal effect on welfare after a reform in the direction of H is given by

$$dW(x, z, \boldsymbol{\iota}) = \partial_x W(x, z, \boldsymbol{\iota}) \left[-H(z) + \left(1 + \frac{\partial_z W(x, z, \boldsymbol{\iota})}{\partial_x W(x, z, \boldsymbol{\iota})} \frac{1}{1 - T'(z)} \right) [1 - T'(z)] dz_{\boldsymbol{\iota}}(T; H) \right] d\mu. \quad (\text{E.2})$$

The first term in (E.2), represents the difference between a dollar in the hands of a household and a dollar in the hands of the government. The second term captures the consequences of behavioral responses for government revenues. Typically, this second term is zero due to the envelope condition. When there is dissonance, however, the envelope arguments normally used to capture the impact on the welfare of these perturbations (e.g., [Saez \(2001\)](#)) must therefore be amended to include this term.²⁸

Let us derive expressions for $\partial_z W(x, z, \boldsymbol{\iota}) / \partial_x W(x, z, \boldsymbol{\iota})$ and $dz_{\boldsymbol{\iota}}(T; h)$. To make our point in as stark a manner as possible and to better communicate with the numerical example of Section 4, we restrict our attention to separable preferences of the form $U(x_i, l_i) = u(x_i) - h(l_i)$ for functions $u : \mathbb{R}_+ \rightarrow \mathbb{R}$ and $h : \mathbb{R}_+ \rightarrow \mathbb{R}$. We then define, with some abuse of notation and omitting parameters for shortness, $x_a(x)$, $x_b(x)$, $z_a(z)$ and $z_b(z)$ as the solution to the family program in (2).²⁹ In this case, we can show that

$$\frac{\partial_z W(x, z, \boldsymbol{\iota})}{\partial_x W(x, z, \boldsymbol{\iota})} \frac{1}{1 - T'(z)} = - \frac{(2\alpha - 1) z'_a(z) + (1 - \alpha)}{(2\alpha - 1) x'_a(x) + (1 - \alpha)}.$$

Of course, if $\alpha = 1/2$, there is no dissonance and the marginal rate of substitution, from the planner's perspective, is equal to the marginal retention rate $1 - T'(z)$. This is also true in the limiting case when $x'_a(x) = z'_a(z)$, but this result is unexpected.³⁰

The behavioral response to a reform in the direction of H , measured by the Gateaux derivative, is given by

$$dz_{\boldsymbol{\iota}}(T; H) = - \frac{\varepsilon^c(z, \boldsymbol{\iota}) z H'(z) + \eta(z, \boldsymbol{\iota}) H(z)}{1 - T'(z) + \varepsilon^c(z, \boldsymbol{\iota}) z T''(z)}, \quad (\text{E.3})$$

where $\varepsilon^c(z, \boldsymbol{\iota})$ is the compensated elasticity of household taxable earnings with respect to the retention rate $1 - T'(z)$, and $\eta(z, \boldsymbol{\iota})$ is the income elasticity of taxable earnings of a type $\boldsymbol{\iota}$, both of which are explicitly calculated in Appendix D.

²⁸It is easy to see the source of these differences. The first-order necessary conditions for program (E.1) is $1 - T'(z) = -\partial_z V(x, z, \boldsymbol{\iota}) / \partial_x V(x, z, \boldsymbol{\iota})$. In a standard Mirrleesian model, $W(x, z, \boldsymbol{\iota}) = V(x, z, \boldsymbol{\iota})$, so that $1 - T'(z) = -\partial_z W(x, z, \boldsymbol{\iota}) / \partial_x W(x, z, \boldsymbol{\iota})$.

²⁹Notice that separability of the utility implies separability between consumption and labor decisions. That is why we write $x_a(\cdot)$ and $x_b(\cdot)$ as a function only of x , and $z_a(\cdot)$ and $z_b(\cdot)$ as function only of z .

³⁰While $x'_a(x)$ depends on the local curvature of $u(\cdot)$ at the optimal choice $(x_a(x), x - x_a(x))$, $z'_a(z)$ depends on the local curvature of $h(\cdot)$ at the optimal choice $(z_a(z), z - z_a(z))$. Hence, they will differ for almost all parametrization.

The relevant elasticities For each household $\boldsymbol{\iota}$, we follow [Jacquet et al. \(2013\)](#) and [Scheuer and Werning \(2017\)](#) in deriving the relevant elasticities of taxable income that take into account the endogeneity of marginal tax rates due to the local tax schedule curvature: uncompensated, $\epsilon(z, \boldsymbol{\iota})$, income, $\eta(z, \boldsymbol{\iota})$, and compensated, $\epsilon^c(z, \boldsymbol{\iota})$ (see Appendix D).

All of these elasticities are defined at the household level. However, the tax schedule should necessarily treat all families with the same total income equally. Therefore, we need to aggregate across all families. For each $z \in \mathbb{R}_+$, let

$$\bar{\varphi}(\boldsymbol{\iota}|z) = \int_{z_{\boldsymbol{\iota}}(T)=z} dF(\boldsymbol{\iota})$$

be the distribution of household types $\boldsymbol{\iota}$ choosing z at the candidate optimal tax schedule T . We can aggregate across households choosing z using $\bar{\varphi}(\boldsymbol{\iota}|z)$ and define aggregate elasticities as follows:

$$\varepsilon(z) = \int_{\boldsymbol{\iota} \in \Lambda} \varepsilon(z, \boldsymbol{\iota}) d\bar{\varphi}(\boldsymbol{\iota}|z), \quad \eta(z) = \int_{\boldsymbol{\iota} \in \Lambda} \eta(z, \boldsymbol{\iota}) d\bar{\varphi}(\boldsymbol{\iota}|z) \quad \text{and} \quad \varepsilon(z)^c = \int_{\boldsymbol{\iota} \in \Lambda} \varepsilon^c(z, \boldsymbol{\iota}) d\bar{\varphi}(\boldsymbol{\iota}|z).$$

Let us denote

$$\Phi(z) = \int_{\boldsymbol{\iota} \in \Lambda} d\bar{\varphi}(\boldsymbol{\iota}|z)$$

as the empirical income distribution induced by the tax system, and let us assume it admits a density, which we shall denote by $\varphi(z)$.

As it turns out, elasticities $\varepsilon(z)$, $\eta(z)$, and $\varepsilon^c(z)$ are important for understanding the consequences of behavioral responses to government revenues. However, a novelty we uncover is that we need alternative elasticities to assess the welfare consequence of behavioral responses. For this, let the average welfare weight at income z , $g(z)$, be defined as

$$g(z) = \int_{\boldsymbol{\iota} \in \Lambda} g(z, \boldsymbol{\iota}) d\bar{\varphi}(\boldsymbol{\iota}|z),$$

where $g(z, \boldsymbol{\iota}) = \partial_x W(z - T(z), z, \boldsymbol{\iota})$, which is positive given our specification of the welfare function (3). Hence, we define the welfare-weighted elasticity by

$$\bar{\varepsilon}(z) = \int_{\boldsymbol{\iota} \in \Lambda} \frac{g(z, \boldsymbol{\iota})}{g(z)} \epsilon(z, \boldsymbol{\iota}) d\bar{\varphi}(\boldsymbol{\iota}|z),$$

with analogous definitions for $\bar{\eta}(z)$, the welfare-weighted income elasticity, and for $\bar{\varepsilon}^c(z)$, the welfare-weighted compensated elasticity of taxable earnings.³¹

³¹Of course, if for all z , $\varepsilon(z, \boldsymbol{\iota})$ is the same for all $\boldsymbol{\iota}$ (or if the co-variance between $\partial W / \partial x$ and ϵ over the distribution of $\boldsymbol{\iota}$ is zero), then the two concepts coincide.

In what follows, it will be convenient to define the household ι *behavioral wedge*:

$$\xi(z, \iota) \equiv \frac{\partial_z W(z - T(z), z, \iota)}{\partial_x W(z - T(z), z, \iota)} \frac{1}{1 - T'(z)}. \quad (\text{E.4})$$

Analogously to what we have done with elasticities, we define

$$\bar{\xi}(z) = \int_{\iota} \frac{g(z, \iota)}{g(z)} \xi(z, \iota) d\bar{\varphi}(\iota|z),$$

the welfare-weighted average behavioral wedge.

E.1 Optimal tax formula

We have already shown how dissonance changes the direct impact of tax reforms on welfare measured by the planner. Unlike the standard case of single households, the welfare impact of a small reform is not negligible. This happens because the social marginal value of income does not coincide with the private one. Adding the mechanical and behavioral effects on tax revenues along the lines of [Saez \(2001\)](#), we arrive at Proposition 5 at the expression for optimal marginal tax rates.

Proposition 5. *The first-order necessary conditions for the optimal joint tax schedule at the total household income level $z \in \mathbb{R}_+$ can be written as*

$$\begin{aligned} \frac{T'(z)}{1 - T'(z)} = & \frac{1}{\epsilon^c(z)} \frac{1 - \Phi(z)}{z\varphi(z)} \left\{ \int_z^{\bar{z}} \left[1 - g(\tilde{z}) + \frac{T'(\tilde{z})}{1 - T'(\tilde{z})} \eta(\tilde{z}) \right] \frac{\varphi(\tilde{z})}{1 - \Phi(z)} d\tilde{z} \right. \\ & + \int_z^{\bar{z}} g(\tilde{z}) \{ [1 - \bar{\xi}(\tilde{z})] \bar{\eta}(\tilde{z}) - \zeta^{\eta, \xi}(\tilde{z}) \} \frac{\varphi(\tilde{z})}{1 - \Phi(z)} d\tilde{z} \Big\} \\ & - \frac{g(z)}{\epsilon^c(z)} \{ [1 - \bar{\xi}(z)] \bar{\epsilon}^c(z) - \zeta^{\epsilon, \xi}(z) \}, \quad (\text{E.5}) \end{aligned}$$

where $\Phi(z)$ is the empirical income distribution induced by the optimal tax system, $\varphi(z)$ is its density, $g(z)$ is the marginal social value of income in the hands of households with income z , $\bar{\xi}(z)$ is the behavioral wedge between the social values of income and its market price, $\bar{\epsilon}^c(z)$ and $\bar{\eta}^c(z)$ are welfare-adjusted average elasticities for households choosing z , and $\zeta^{\epsilon, \xi}(z)$ and $\zeta^{\eta, \xi}(z)$ are the welfare-adjusted covariances between these elasticities and the behavioral wedges for all households choosing z .

The first thing to note about (E.5) is that if $\bar{\xi}(z) = 1$, this is simply the [Piketty-Saez](#) intuitive expression for optimal taxes. The second important thing is that two types of aggregation are used to characterize optimal taxes: the regular empirical frequency-weighted

elasticities ($\varepsilon(z)$, $\varepsilon^c(z)$, and $\eta(z)$) and the welfare-weighted elasticities ($\bar{\varepsilon}(z)$, $\bar{\varepsilon}^c(z)$, and $\bar{\eta}(z)$).

As pointed out by [Saez \(2001\)](#), using a perturbation approach allows us to write optimal tax formulae even when types are multidimensional, which is precisely our case here. However, with multidimensional heterogeneity, households of different types bunch at the same z , and we need to aggregate behavioral responses across these agents.³² This is already in [Saez \(2001\)](#), as we can see if we assume $\bar{\xi}(z) = 1$. The novelty is that, whereas elasticities are only important in [Saez \(2001\)](#) because of their behavioral impact on revenues, they directly affect utility as perceived by the planner. In this case, averages must be welfare-adjusted analogously to risk adjustments in asset pricing formulae.

Dissonance means that behavioral responses have first-order consequences on welfare, which vary across agents with the same earnings. The overall impact, which depends on the elasticities and the wedges, is weighted not by the relative empirical frequency of couples choosing z but by the welfare-adjusted frequencies. All elasticities pertaining to dissonance have a bar to denote this adjustment. The covariance terms that appear in the last two lines of (E.5) are also calculated under this welfare-adjusted measure.³³

As it turns out, ridding the expression of these aggregation issues leads to a considerably more straightforward extension of optimal tax formulae. Toward this end, we assume that all elasticities are identical for agents choosing the same z . This will lead us to an ABC+D formula (Proposition 4) that is derived from (E.6) below.

Corollary 2. *Assume that for all z , $\varepsilon(z, \iota) = \varepsilon(z)$, $\eta(z, \iota) = \eta(z)$ and $\varepsilon^c(z, \iota) = \varepsilon^c(z)$ for all ι choosing z . Then, the first-order condition for the optimal joint tax schedule for couples at income level z in Proposition 5 can be rewritten as*

$$\begin{aligned} \frac{T'(z)}{1 - T'(z)} = \frac{1}{\varepsilon^c(z)} \frac{1 - \Phi(z)}{z\varphi(z)} \left\{ \int_z^{\bar{z}} \left[1 - g(\tilde{z}) + \frac{T'(\tilde{z})}{1 - T'(\tilde{z})} \eta(\tilde{z}) \right] \frac{\varphi(\tilde{z})}{1 - \Phi(z)} d\tilde{z} \right. \\ \left. + \int_z^{\bar{z}} g(\tilde{z}) [1 - \bar{\xi}(\tilde{z})] \eta(\tilde{z}) \frac{\varphi(\tilde{z})}{1 - \Phi(z)} d\tilde{z} \right\} - \bar{g}(z) [1 - \bar{\xi}(z)]. \quad (\text{E.6}) \end{aligned}$$

Because we have assumed that elasticities are identical for all households choosing the same z , we can drop the bar from almost all statistics and eliminate the covariance terms in (E.5). The only term for which we still make a distinction between the empirical

³²According to [Saez \(2001\)](#) "The direct proof using elasticities shows that it is not necessary to introduce a uni-dimensional exogenous skill distribution to obtain formula (14) [our formula (E.5)]. Therefore, formula (14) might, in principle, be valid for any heterogeneous population as long as (...) are considered as average elasticities at income level z ."

³³Covariance terms also appear in the optimal tax formulae for behavioral agents derived by [Gerritsen \(2016\)](#). [Gerritsen](#)'s representation slightly differs from ours since he does not work with a change in the measure, considering extra co-variance terms in the empirical measure.

distribution and the welfare-adjusted distribution concerns the behavioral wedge $\bar{\xi}(z)$.

Note how (E.6) identifies a new sufficient statistic that should be considered when evaluating optimal income taxes. It highlights that the planner may want to impose additional distortions to influence decisions taken within the household.

Equation (E.6) defines $T'(z)/(1 - T'(z))$ implicitly. We can find an explicit solution by solving this integral equation. This is the content of Proposition 4 in the main text.

Relationship between the optimal taxation formulae Optimal tax formulae (23) could be directly derived from (E.5). First, note that under identical $\ln$ preferences, the marginal social value of resources is equal for all households earning the same z , leading us to drop all of the covariance terms and equalize the empirical and welfare-adjusted moments: the relevant expression for optimal taxes takes the explicit form (25). Second, separability greatly simplifies the $C(z)$ term in (25) once one notes that, under separability, $\eta(z)/\varepsilon^c(z) = d \ln u'(z - T(z))/dz$. A somewhat tedious algebra is needed to show that the dissonance term, $D(z)$, becomes the term in the second line of (23). As in Saez (2001), it is comforting to check that the same formulae, albeit written in terms of primitives only, are reached using a mechanism design approach.

Proof of Proposition 5. The Gateaux derivative of the earning supply functional is defined by

$$dz_{\iota}(T; H) \equiv \lim_{\mu \rightarrow 0} \frac{z_{\iota}(T + \mu H) - z_{\iota}(T)}{\mu} = \left. \frac{\partial z_{\iota}}{\partial \mu} \right|_{\mu=0}. \quad (\text{E.7})$$

Increase the marginal tax rate by $d\tau$ in a small interval $(z', z' + dz')$. Let $\Phi(z)$ denote the distribution of income induced by the candidate optimal tax schedule, $T(\cdot)$ (and the distribution $F(\cdot)$ of types ι). Let us then consider the overall impact of such a reform.

Impact on government revenues For agents with taxable income in the interval $(z', z' + dz')$, there is a mechanical effect $M_1 = d\tau dz'$. But there is also a behavioral effect, which is given by

$$B_1(z', \iota) = T'(z') \left. \frac{dz}{d\tau} \right|_u d\tau = T'(z') \epsilon^c(z', \iota) \frac{z'}{1 - T'(z')} d\tau.$$

Let $\bar{\varphi}(\iota|z')$ be the conditional distribution on z' of ι . Then, the average response at income z' is

$$B_1(z') = \frac{z' T'(z')}{1 - T'(z')} \int \epsilon^c(z', \iota) d\bar{\varphi}(\iota|z') d\tau = \frac{z' T'(z')}{1 - T'(z')} \epsilon^c(z') d\tau,$$

for $\epsilon^c(z')$ as defined in Section E.

This affects agents in the interval $(z', z' + dz')$, which results in an overall effect

$$\epsilon^c(z') \frac{T'(z')}{1 - T'(z')} z' \varphi(z') dz' d\tau,$$

where φ is the density of z induced by the tax system.

For agents with taxable income in $[z', \bar{z}]$, we have again a mechanical effect $d\tau dz'$ which affects all agents for an overall increase in revenues given by

$$M_2 = [1 - \Phi(z')] d\tau dz'.$$

For all agents in this interval, there is also a behavioral response due to the income effect

$$B_2(z, \mathbf{u}) = T'(z) \frac{dz}{dI} d\tau dz' = \frac{T'(z)}{1 - T'(z)} \eta(z, \mathbf{u}) d\tau dz',$$

which can be aggregated across all households earning z through

$$B_2(z) = \frac{T'(z)}{1 - T'(z)} \int \eta(z, \mathbf{u}) d\bar{\varphi}(\mathbf{u}|z) d\tau dz' = \frac{T'(z)}{1 - T'(z)} \eta(z) d\tau dz'.$$

Finally, we can use the expression above to find the overall behavior effect on tax revenues for agents in this range

$$B_2(z, \mathbf{u}) = d\tau dz' \int_{z'}^{\bar{z}} \frac{T'(z)}{1 - T'(z)} \eta(z) \varphi(z) dz.$$

This is exactly as in [Saez \(2001\)](#) with the proviso that we are using a different definition for the elasticities we are aggregating.

Welfare impacts As for the welfare effects, again, let us start with households with taxable income in the interval $(z', z' + dz')$. First note that for this group, the tax reform has no first-order mechanical effect on utility, V . The relevant ‘decision’ elasticities are, therefore, Hicksian.

The impact on welfare is however measured according to the planner’s valuation, i.e., the experience’ utility W . Hence,

$$dW_1(z', \mathbf{u}) = \frac{\partial}{\partial x} W(z' - T(z'), z', \mathbf{u}) [1 - T'(z')] \frac{dz}{d\tau} \Big|_u d\tau + \frac{\partial}{\partial z} W(z' - T(z'), z', \mathbf{u}) \frac{dz}{d\tau} \Big|_u d\tau,$$

which simplifies to

$$dW_1(z', \boldsymbol{\iota}) = \frac{\partial}{\partial x} W(z' - T(z'), z', \boldsymbol{\iota}) \left[1 - T'(z') + \frac{\frac{\partial}{\partial z} W(z' - T(z'), z', \boldsymbol{\iota})}{\frac{\partial}{\partial x} W(z' - T(z'), z', \boldsymbol{\iota})} \right] \frac{dz}{d\tau} \Big|_u d\tau,$$

or

$$dW_1(z', \boldsymbol{\iota}) = \frac{\partial}{\partial x} W(z' - T(z'), z', \boldsymbol{\iota}) [1 - \xi(z', \boldsymbol{\iota})] \epsilon^c(z', \boldsymbol{\iota}) z' d\tau,$$

where for all z ,

$$\xi(z, \boldsymbol{\iota}) \equiv \frac{1}{1 - T'(z)} \frac{\frac{\partial}{\partial z} W(z - T(z), z, \boldsymbol{\iota})}{\frac{\partial}{\partial x} W(z - T(z), z, \boldsymbol{\iota})}.$$

Let $\bar{\varphi}(\boldsymbol{\iota}|z')$ the conditional on z' distribution of $\boldsymbol{\iota}$. The overall welfare effect at z' is

$$W_1 = dz' d\tau z' \int [g(z', \boldsymbol{\iota}) [1 - \xi(z', \boldsymbol{\iota})] \epsilon^c(z', \boldsymbol{\iota})] d\bar{\varphi}(\boldsymbol{\iota}|z'),$$

for $g(z', \boldsymbol{\iota})$ the marginal social value of income in the hands of a $\boldsymbol{\iota}$ household earning z' .

To simplify this expression, let $\tilde{\varphi}(\cdot|z)$ be a measure equivalent to $\bar{\varphi}(\cdot|z)$, ($\tilde{\varphi}(\cdot|z) \equiv \bar{\varphi}(\cdot|z)$) whose Radon-Nikodym derivative with respect to $\bar{\varphi}(\cdot|z')$ is the function $g(z, \cdot)/\bar{g}(z)$, where $\bar{g}(z) = \int g(z, \boldsymbol{\iota}) d\bar{\varphi}(\boldsymbol{\iota}|z)$. This measure allows us to re-express the definitions for $\bar{\xi}(z)$ and $\bar{\epsilon}^c(z)$ from Appendix E, as $\bar{\xi}(z) = \int \xi(z, \boldsymbol{\iota}) d\tilde{\varphi}(\boldsymbol{\iota}|z) = \tilde{\mathbb{E}}[\xi(z, \boldsymbol{\iota})]$, and $\bar{\epsilon}^c(z) = \tilde{\mathbb{E}}[\epsilon^c(z, \boldsymbol{\iota})]$. It also allows us to define a covariance term

$$\zeta^{\epsilon, \xi}(z) = \text{cov}(\epsilon^c(z, \boldsymbol{\iota}), \xi(z, \boldsymbol{\iota})) = \tilde{\mathbb{E}}[\epsilon^c(z, \boldsymbol{\iota}) \xi(z, \boldsymbol{\iota})] - \bar{\xi}(z) \bar{\epsilon}^c(z).$$

Using these definitions we write

$$W_1 = dz' d\tau z' \bar{g}(z') [[1 - \bar{\xi}(z')] \bar{\epsilon}^c(z') - \zeta^{\epsilon, \xi}(z')].$$

For households with taxable income $z \geq z'$,

$$dW_2(z, \boldsymbol{\iota}) = -\frac{\partial}{\partial x} W(z - T(z), z, \boldsymbol{\iota}) dz' d\tau + \frac{\partial}{\partial x} W(z - T(z), z, \boldsymbol{\iota}) [1 - T'(z)] \frac{dz}{dI} dz' d\tau + \frac{\partial}{\partial z} W(z - T(z), z, \boldsymbol{\iota}) \frac{dz}{dI} dz' d\tau$$

leading to

$$dW_2(z, \boldsymbol{\iota}) = -\frac{\partial}{\partial x} W(z - T(z), z, \boldsymbol{\iota}) \left\{ 1 + \left[1 - T'(z) + \frac{\frac{\partial}{\partial z} W(z - T(z), z, \boldsymbol{\iota})}{\frac{\partial}{\partial x} W(z - T(z), z, \boldsymbol{\iota})} \right] \frac{dz}{dI} \right\} dz' d\tau$$

and finally

$$dW_2(z, \boldsymbol{\iota}) = -g(z, \boldsymbol{\iota}) \{1 + [1 - \xi(z, \boldsymbol{\iota})] \eta(z, \boldsymbol{\iota})\} dz' d\tau.$$

To aggregate across all household earnings at the same z , we integrate over $\boldsymbol{\iota}$

$$d\bar{W}_2(z) = dz' d\tau \int \left[-g(z, \boldsymbol{\iota}) \{1 + [1 - \xi(z, \boldsymbol{\iota})] \eta(z, \boldsymbol{\iota})\} \right] d\bar{\varphi}(\boldsymbol{\iota}|z').$$

Analogously, to what we have done for dW_1 we can further simplify this expression to

$$d\bar{W}_2(z) = d\tau dz' \bar{g}(z) \left[[1 - \bar{\xi}(z)] \bar{\eta}^c(z) - \zeta^{\eta, \xi}(z) \right],$$

where the covariance term $\zeta^{\eta, \xi}(z)$ is also under the equivalent measure, $\tilde{\varphi}(\cdot|z)$.

The overall effect on this household welfare is

$$W_2 = d\tau dz' \int_{z'}^{\bar{z}} \bar{g}(z) \{1 + [1 - \bar{\xi}(z)] \bar{\eta}(z) - \zeta^{\eta, \xi}(z)\} \varphi(z) dz.$$

At the optimum, all these effects must cancel out and we get

$$M_1 + M_2 + B_1 + B_2 + W_1 + W_2 = 0.$$

Substituting the expressions for each one of these terms we get

$$\begin{aligned} & \frac{T'(z')}{1 - T'(z')} \varepsilon^c(z') z' \varphi(z') - g(z') \left[[1 - \xi(z')] \bar{\varepsilon}^c(z') - \zeta^{\varepsilon, \xi}(z') \right] z' \varphi(z') + [1 - \Phi(z')] \\ & + \int_{z'}^{\bar{z}} \frac{T'(z)}{1 - T'(z)} \eta(z) \varphi(z) dz + \int_{z'}^{\bar{z}} g(z) \{1 + [1 - \xi(z)] \bar{\eta}(z) - \zeta^{\eta, \xi}(z)\} \varphi(z) dz = 0. \end{aligned}$$

Reorganizing the terms,

$$\begin{aligned} \frac{T'(z')}{1 - T'(z')} \varepsilon^c(z') z' \varphi(z') &= \int_{z'}^{\bar{z}} \left[1 - g(z) + \frac{T'(z)}{1 - T'(z)} \eta(z) \right] \varphi(z) dz \\ &+ \int_{z'}^{\bar{z}} g(z) \{ [1 - \xi(z)] \bar{\eta}(z) - \zeta^{\eta, \xi}(z) \} \varphi(z) dz \\ &+ g(z') \left[[1 - \xi(z')] \bar{\varepsilon}^c(z') - \zeta^{\varepsilon, \xi}(z') \right] z' \varphi(z') \end{aligned}$$

or

$$\begin{aligned} \frac{T'(z)}{1-T'(z)} &= \frac{1-\Phi(z)}{z\varphi(z)\varepsilon^c(z)} \left\{ \int_z^{\bar{z}} \left[1 - g(\tilde{z}) + \frac{T'(\tilde{z})}{1-T'(\tilde{z})} \eta(\tilde{z}) \right] \frac{\varphi(\tilde{z})}{1-\Phi(z)} dz \right. \\ &\quad \left. + \int_z^{\bar{z}} g(\tilde{z}) \{ [1 - \xi(\tilde{z})] \bar{\eta}(\tilde{z}) - \zeta^{\eta, \xi}(\tilde{z}) \} \frac{\varphi(\tilde{z})}{1-\Phi(z)} dz \right\} \\ &\quad - \frac{g(z)}{\varepsilon^c(z)} \{ [1 - \xi(z)] \bar{\varepsilon}^c(z) - \zeta^{\varepsilon, \xi}(z) \}. \quad (\text{E.8}) \end{aligned}$$

□

Proof of Proposition 2. We prove Proposition 2 as a particular case of Proposition 5. Under the assumptions of Proposition 2, $\xi(z) = \bar{\xi}(z)$, $\varepsilon^c(z) = \bar{\varepsilon}^c(z)$ and $\eta(z) = \bar{\eta}(z)$, for all z . Moreover, $\zeta^{\varepsilon, \xi}(z) = 0$ and $\zeta^{\eta, \xi}(\tilde{z}) = 0$, for all z . As a consequence, expression (E.5) collapses to

$$\begin{aligned} \frac{T'(z)}{1-T'(z)} &= \frac{1-\Phi(z)}{z\varphi(z)\varepsilon^c(z)} \left\{ \int_z^{\bar{z}} \left[1 - g(\tilde{z}) + \frac{T'(\tilde{z})}{1-T'(\tilde{z})} \eta(\tilde{z}) \right] \frac{\varphi(\tilde{z})}{1-\Phi(z)} dz \right. \\ &\quad \left. + \int_z^{\bar{z}} g(\tilde{z}) \{ [1 - \xi(\tilde{z})] \bar{\eta}(\tilde{z}) \} \frac{\varphi(\tilde{z})}{1-\Phi(z)} dz \right\} \\ &\quad - \frac{g(z)}{\varepsilon^c(z)} \{ [1 - \xi(z)] \bar{\varepsilon}^c(z) \}, \quad (\text{E.9}) \end{aligned}$$

or

$$\begin{aligned} \frac{T'(z)}{1-T'(z)} &= \frac{1}{z\varphi(z)\varepsilon^c(z)} \int_z^{\bar{z}} \left[1 - g(\tilde{z}) + g(\tilde{z}) [1 - \xi(\tilde{z})] \bar{\eta}(\tilde{z}) + \frac{T'(\tilde{z})}{1-T'(\tilde{z})} \eta(\tilde{z}) \right] \varphi(\tilde{z}) d\tilde{z} \\ &\quad - g(z) [1 - \xi(z)]. \quad (\text{E.10}) \end{aligned}$$

If we define

$$\kappa(z) := \frac{T'(z)}{1-T'(z)},$$

$$A(z) := \varepsilon^c(z) z \varphi(z),$$

$$B(\tilde{z}) := \{ 1 - g(\tilde{z}) + g(\tilde{z}) [1 - \xi(\tilde{z})] \bar{\eta}(\tilde{z}) \} \varphi(\tilde{z}),$$

$$C(\tilde{z}) := \eta(\tilde{z}) \varphi(\tilde{z})$$

and

$$D(z) = g(z) [1 - \xi(z)] A(z),$$

then (E.8) can be written as

$$A(z)\kappa(z) = \int_z^{\bar{z}} [B(\tilde{z}) + C(\tilde{z})\kappa(\tilde{z})] d\tilde{z} - D(z)$$

and solved for

$$\begin{aligned} \kappa(z) = & -\frac{1}{A(z)} \int_z^{\bar{z}} \left[B(\tilde{z}) - \frac{D(\tilde{z})C(\tilde{z})}{A(\tilde{z})} \right] \exp \left\{ \int_z^{\tilde{z}} \frac{C(\tilde{\tilde{z}})}{A(\tilde{\tilde{z}})} d\tilde{\tilde{z}} \right\} d\tilde{z} \\ & + \frac{D(z)}{A(z)} - \frac{D(\bar{z})}{A(\bar{z})} \exp \left\{ \int_z^{\bar{z}} \frac{C(\tilde{z})}{A(\tilde{z})} d\tilde{z} \right\}. \end{aligned}$$

Hence,

$$\begin{aligned} \frac{T'(z)}{1 - T'(z)} = & \frac{1}{\varepsilon^c(z)} \frac{1 - \Phi(z)}{\varphi(z)z} \int_z^{\bar{z}} \{g(\tilde{z}) - 1\} \exp \left\{ \int_{\tilde{z}}^z \frac{\eta(\tilde{\tilde{z}})}{\varepsilon^c(\tilde{\tilde{z}})} \frac{d\tilde{\tilde{z}}}{\tilde{\tilde{z}}} d\tilde{\tilde{z}} \right\} \frac{\varphi(\tilde{z})}{1 - \Phi(z)} d\tilde{z} \\ & - g(z) \{1 - \xi(z)\}. \quad (\text{E.11}) \end{aligned}$$

□

F Gender-based Taxes

We consider the impact of a reform that increases w_a , i.e., $\hat{w}_a = w_a(1 - t)$ for $t < 0$ while adjusting $T(\cdot)$ to hold (z, x) fixed.

The household optimal choice of z_a and z_b to attain earnings z is

$$\min_{z_a \in [0, z]} \left\{ \alpha \left(\frac{z_a}{w_a} \right)^{1+\gamma} + (1 - \alpha) \left(\frac{z - z_a}{w_b} \right)^{1+\gamma} \right\}$$

and the first-order condition is

$$\alpha \frac{z_a^\gamma}{w_a^{1+\gamma}} - (1 - \alpha) \frac{(z - z_a)^\gamma}{w_b^{1+\gamma}} = 0.$$

Now consider a reform that increases w_a but changes $T(\cdot)$ in such a way as to keep (z, x) fixed. In this case,

$$\left[\gamma \alpha \frac{z_a^{\gamma-1}}{w_a^{1+\gamma}} + \gamma (1 - \alpha) \frac{(z - z_a)^{\gamma-1}}{w_b^{1+\gamma}} \right] dz_a - \alpha (1 + \gamma) \frac{z_a^\gamma}{w_a^{2+\gamma}} dw_a = 0$$

or

$$\left[\frac{z_a^\gamma}{w_a^{1+\gamma}} + \frac{1-\alpha}{\alpha} \frac{z_b^\gamma}{w_b^{1+\gamma}} \frac{z_a}{z_b} \right] \frac{dz_a}{z_a} - \frac{1+\gamma}{\gamma} \frac{z_a^\gamma}{w_a^{1+\gamma}} \frac{dw_a}{w_a} = 0.$$

Using the first-order condition,

$$\left[\frac{z_a^\gamma}{w_a^{1+\gamma}} + \frac{z_a^\gamma}{w_a^{1+\gamma}} \frac{z_a}{z_b} \right] \frac{dz_a}{z_a} - \frac{1+\gamma}{\gamma} \frac{z_a^\gamma}{w_a^{1+\gamma}} \frac{dw_a}{w_a} = 0,$$

which can be simplified to

$$\left[1 + \frac{z_a}{z_b} \right] \frac{dz_a}{z_a} - \frac{1+\gamma}{\gamma} \frac{dw_a}{w_a} = 0.$$

Hence,

$$\frac{dz_a}{z_a} = [1 - k_a] \frac{1+\gamma}{\gamma} \frac{dw_a}{w_a}.$$

An increase in w_a leads to an increase in l_a whenever $k_a < 1/(1+\gamma)$. We have that

$$\left. \frac{dV_a}{dw_a} \right|_z = -l_a^\gamma \left. \frac{dl_a}{dw_a} \right|_z = - \left[[1 - k_a] \frac{1+\gamma}{\gamma} - 1 \right] l_a^{1+\gamma} \frac{dw_a}{w_a}$$

In this case,

$$\left. \frac{dV_a}{dw_a} \right|_z < 0 \iff 1 - k_a > \frac{\gamma}{1+\gamma}.$$

G Dissonance Heuristics

In Subsection 3.2 we have explained how the misalignment between the planner's and the household's objective will cause the planner to desire the household to increase or decrease earnings relative to the household's choice. The purpose of the discussion below is to explain how this interacts with the inter-household incentive provision to define the optimum.

Consider a simpler economy, that has only two types of households. What we call the [Mirrlees'](#) solution is the one that uses the households' preferences as a normative criterion. Household preferences are represented by the purple indifference curves V_0 , for the low-productivity family, and V_1 , for the high-productivity one. The pair of earnings-consumption bundles, (z, x) , that comprise it are depicted in Figure 6 at point A which is the intersection of purple indifference curves V_0 and V_1 (for high productivity households) and at point B in V_1 which has the slope of the budget line, i.e., 45° line. At A the high-productivity household is indifferent between its bundle and the low-productivity households' one. The low-productivity household's indifference curve has a lower than

45° slope with downward distorted earnings.

At the *Mirrlees*'s optimum, the household welfare according to the planner is represented by the gray indifference curves, W_0 for the low-productivity household, and W_1 for the high-productivity household. The steeper indifference curves corresponds to the case $\xi > 1$. The area to the southwest of the low-productivity household's bundle above the W_0 and below the V_0 curve represents all the bundles that are deemed better for the low-productivity household by the planner, but not by the household. For this large area in the figure, they are also incentive-compatible since they are below V_1 . As for feasibility, all the bundles below the 45° line crossing point A are also feasible. Hence, the dashed area between the W_0 indifference curve and the highest 45° line defines the set of feasible bundles that the planner considers better for the low-productivity household than the standard *Mirrlees*'s optimal bundle. The *Better than Mirrlees* bundle depicted along the same 45° line is also incentive compatible as made clear by the fact that the high-productivity household's bundle lies in the lower contour set bounded by $\hat{V}_0$. There are even better incentive-feasible allocations, according to the planner's objective.

The configuration of the indifference curves is particular both in that we could have assumed different directions of dissonance and in the fact that at the *Mirrlees*' solution, there is a region of superior (according to the planner) incentive-feasible allocations.³⁴ In general, a reform from a *Mirrlees*' optimum is more involved but is almost always possible.

³⁴This configuration corresponds to the case in which the dissonance index, as defined in (7) is greater than one. The case in which it is less than one is equally plausible from a theoretical perspective.

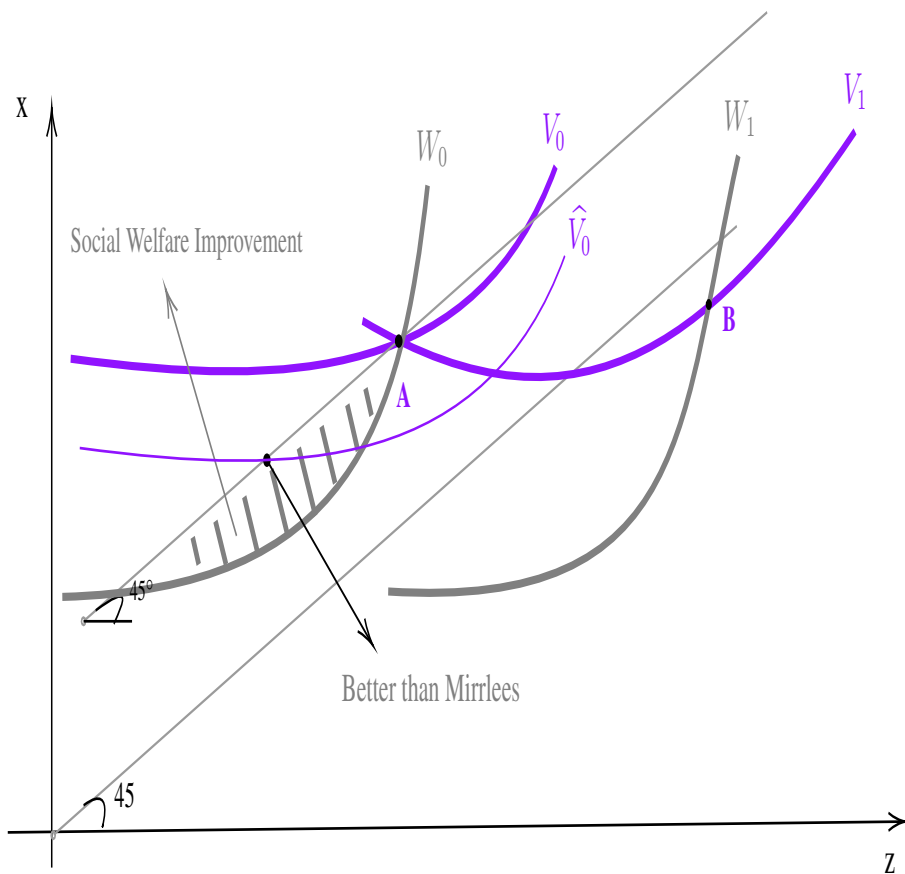


Figure 6: The standard [Mirrlees](#)'s solution is the pair of bundles A , B . The gray indifference curves represent the planner's assessment of bundles for each household. Here, dissonance leads to steeper indifference curves for the planner, meaning that the planner would prefer households to work less than what they do in the standard [Mirrlees](#)'s solution.